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PAPER

An exploration of enhanced characteristics and efficiency of dye-doped L-Threonine single crystal

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E-mail: www_saravanan@yahoo.com**Keywords:** FTIR, second harmonic generation, piezoelectric study, LDT, nonlinear optical material

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Abstract

The improved crystalline characteristics of a crystal doped with 0.5 mol% methylene blue dye was analyzed and compared using several techniques such as SXRD, FTIR, (Spectrum of absorption and Spectrum of optical transmittance), PXRD, and Second Harmonic Generation (SHG). The crystal was prepared using the slow evaporation method. The work hardening coefficient was evaluated for our synthesized crystal, indicating its soft nature. Additionally, we want to achieve similar levels of second harmonic Generation in both pure L-threonine materials and dye-doped crystals. Research on photoluminescence reveals that fully developed crystals have a reduced number of imperfections. Piezoelectric coefficients (d_{33}) were also calculated for the as-grown Pure L-threonine crystal and L-threonine doped with dye Single crystal. The increased LDT value be attributed to the significantly decreased section, confinement, and mechanical stiffness of both the Pure L-threonine and dye-doped L-threonine Single crystal. The etching tests shown that both the as-grown Pure L-threonine crystal and the dye-doped L-threonine Single crystal had a better level of crystalline fineness and a smaller number of defects. These characteristics are essential for their use in nonlinear optics (NLO) applications.

1. Introduction

L-threonine is a very suitable material for use in photonic applications. The use of several organic dyes in the formation of L-threonine crystals at room temperature resulted in an improvement in second harmonic generation (SHG) efficiency, thanks to the improved optical properties and growth characteristics of the crystals [1]. L-threonine is an amino acid that has both positive and negative charges, making it zwitterionic. It forms crystals that do not have a centre of symmetry. The applicability of space group has expanded to include areas such as optical band pass filters, signal processing, and optical communications [2]. The efficacy of dye-infused crystals for diverse applications such as lasers and optoelectric devices has been shown [3]. A high-quality crystal of L-threonine including methylene blue dye was successfully produced and reported for the first time. Methylene blue, a kind of organic dye with the chemical formula ($C_{16}H_{18}N_3SCl$), is used in the production of laser devices [4, 5]. The presence of a dye integrated into the parent crystal enhances the interaction between photons and electrons, creating an environment that promotes a higher level of second harmonic Generation compared to the pure material. Furthermore, it has an impact on nucleation kinetics, morphology, growth rate, LTD (low temperature degradation), mechanical characteristics, and so on [6]. The following studies have recently evaluated single crystals of the amino acid L-threonine: L-threonine ammonium bromide, which was developed on a single crystal [7]; L-threonine and Nd: L-threonine crystals [8] and L-threonine amino acid crystal [9]. A study has been conducted on the use of organic dye as an addition in an L-threonine host [10]. However, there is currently no existing literature on the incorporation of methylene blue. The saturation and reverse saturation absorption dyes play a vital part in technological improvement due to their outstanding fluorescence capabilities, which are characterised by their vivid and vibrant hues. A high-quality single crystal of

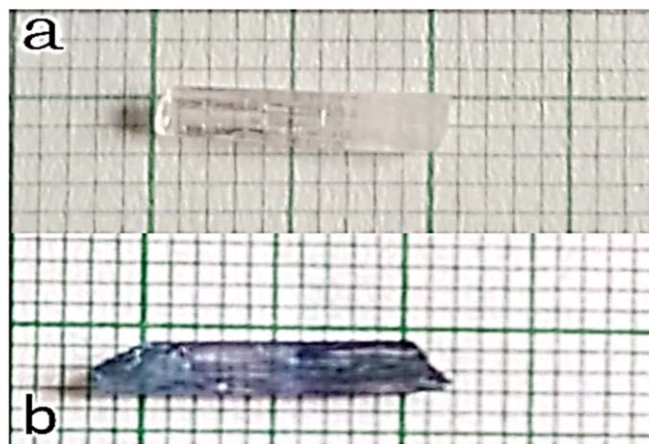


Figure 1. (a) Pure L-threonine and (b) dye doped L-threonine Crystals by SEST method.

L-threonine doped with 0.5% methylene blue dye was successfully generated at room temperature using the slow evaporation method. L-threonine crystal doped with Eu_3^+ [11], L-threonine crystal doped Ce_3^+ [3], Zinc acetate doped L-threonine crystal [12], N-Benzoyl glycine doped L-Threonine [1]. The crystal was then subjected to various characterization techniques including FTIR, UV-vis, PXRD, Etching, LTD, SHG, Morphology, Piezoelectric Effect, DTA/TG, and Mechanical properties analysis.

The incorporation of methylene blue dye into L-threonine single crystals has several uses and can result in diverse applications. Here are some justifications for doing this doping process:

1.1. Enhanced optical properties

Methylene blue dye is renowned for its robust absorption and fluorescence characteristics throughout the visible range. By introducing this dye into L-threonine single crystals, the optical characteristics of the crystal can be augmented, which can result in higher sensitivity and selectivity for optical sensing applications.

1.2. Tunable properties

Doping enables precise adjustment of the optical characteristics of the L-threonine crystal. The absorption and emission spectra of the crystal can be changed to satisfy particular needs for different applications, such as fluorescence imaging or sensing, by changing the quantity of methylene blue dye.

1.3. Functionalization

The addition of methylene blue dye can impart particular functionalities to the L-threonine crystal. For example, the dye can enhance the ability to interact with certain molecules or ions, so making the crystal well-suited for applications in chemical sensing or environmental monitoring [13].

1.4. Optoelectronic devices

By introducing methylene blue dye onto L-threonine single crystals, it becomes possible to create optoelectronic devices like light-emitting diodes (LEDs) or photodetectors. The use of dye doping can increase the optical characteristics of these devices, leading to improved performance and efficiency.

In summary, the addition of methylene blue dye to L-threonine single crystals provides a means to customise their optical and functional characteristics for many purposes, including sensing, imaging, optoelectronics devices.

2. Experimental method

2.1. Crystal growth by SEST method

Single crystals of Pure L-threonine (Sigma Aldrich 98% purity) and 0.5 mol% methylene blue (Sigma Aldrich 99.5%) dye doped L-threonine crystals were developed using the Slow Evaporation Solution Growth (SEST) method at room temperature in an aqueous medium (distilled water). Both mixes become supersaturated after being stirred with a magnetic stirrer for 3 h, however they still look as a homogenous solution. The process of repeatedly purifying and forming large crystals over a 30-day period is known as repeated recrystallization, which ensures high-quality crystals. The solubility and crystallisation of L-threonine in water as a solvent are

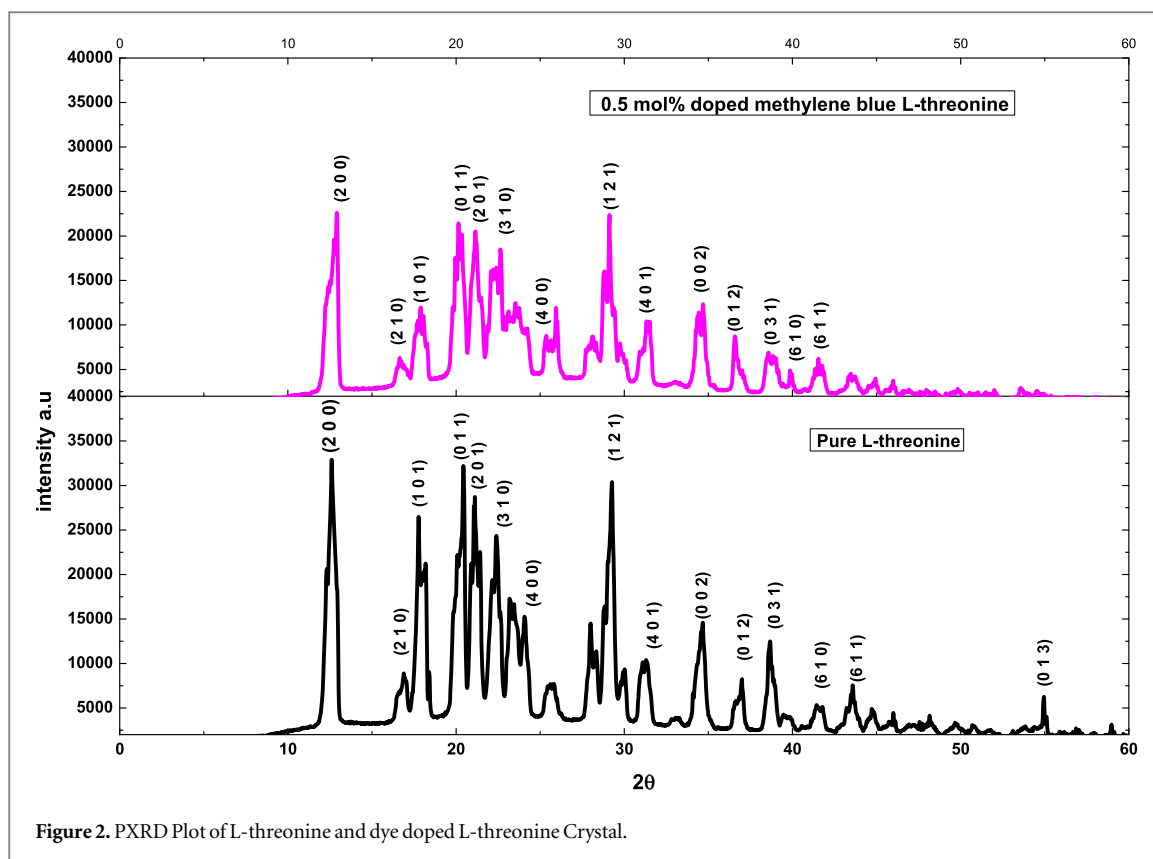


Figure 2. PXRD Plot of L-threonine and dye doped L-threonine Crystal.

Table 1. Single crystal XRD analysis of dye doped L-threonine Crystal.

Unit cell parameters	Pure L-threonine	0.5 mol% of methylene blue doped L-threonine
a(Å)	13.611 Å	13.79482 Å
b(Å)	7.738 Å	7.64002 Å
c(Å)	5.144 Å	5.15281 Å
Volume	541.565(Å) ³	543.0682 (Å) ³

determined by its polar character. Both Pure and dye-doped environment experience the same crystallisation process. The crystallisation process is repeated at various temperatures due to the relationship between solubility and concentration of solute per 100 ml. The solubility rate is modified by the adhesive connection that occurs between Pure L-threonine and the dopant methylene blue. Figures 1(a) and (b) illustrates a pure single crystal and a single crystal doped with dye. Additionally, the dopant is clearly seen to be present inside the parent crystal.

3. Results and discussion

3.1. Single crystal XRD

Single crystal diffraction research was conducted to examine cell characteristics and diffraction patterns of single crystals. The process of forming L-threonine crystals and dye-doped L-threonine crystals take place in an orthorhombic crystal system with non-centrosymmetric space groups P_{212121} . This is confirmed by the lattice parameter values obtained, as given in table 1. The existence of H^+ and $C_4H_8NO_3^-$ zwitterion inside the unit cell leads to the formation of a crystalline structure composed of pure and dye-doped L-threonine, culminating in an orthorhombic lattice [14]. Compared to the pure crystal, the presence of 0.5 mol percent of dye-doped L-threonine resulted in marginally enhanced unit cell parameters. This improvement can be attributed to the stress created in the crystal network by the dye molecules.

Methylene blue molecules could possibly occupy the empty spaces between the atoms in the crystal structure of L-threonine. The occupation of interstitial spaces causes the crystal structure to expand, leading to an increase in volume. The presence of water molecules or other solvents in the crystal lattice of methylene blue can lead to the swelling or expansion of the crystal due to hydration. The process of hydration lead to an augmentation in

Table 2. Powder XRD analysis of dye doped L-threonine Crystal.

Unit cell parameters	Pure L-threonine	0.5 mol% of methylene blue doped L-threonine
Crystal system	Orthorhombic	Orthorhombic
Space Group	P_{212121}	P_{212121}
Unit Cell Dimensions	$a = 13.611 \text{ \AA}$, $b = 7.738 \text{ \AA}$, $c = 5.144 \text{ \AA}$	$a = 13.79482 \text{ \AA}$, $b = 7.64002 \text{ \AA}$, $c = 5.15281 \text{ \AA}$
Volume	$541.565 (\text{\AA})^3$	$543.0682 (\text{\AA})^3$

the size of the crystal. Incorporating methylene blue molecules into the crystal lattice induces stress and strain within the lattice structure. The stress induces the lattice to undergo expansion in order to accommodate the dopant molecules, resulting in a net increase in the volume of the crystal. The addition of methylene blue during crystal development strongly impacts the nucleation and growth mechanisms, possibly leading to the formation of bigger crystals and hence an increase in volume compared to crystals that are not doped. It is crucial to acknowledge that the precise processes that cause the increase in crystal volume could vary depending on many aspects, such as the concentration of dopant molecules, the crystal structure of L-threonine, and the experimental settings during the doping process. Advanced analytical and characterization methods, like as x-ray diffraction or microscopy, used to examine and verify the changes in crystal volume caused by the addition of methylene blue.

3.2. Powder XRD of pure and dye incorporated single crystal

Analysed utilising a Philips pert PRO-Xray diffractometer, the powder sample was examined in both its pure state and when doped with dye in a single crystal. An analysis of the intensity versus 2θ spectrum was conducted. Additionally, the measurement of the PXRD spectrum determines the degree of purity of the phases and the crystalline characteristics of the sample. Fine powder sample preparation has been done using a mortar and pestle to analyse PXRD Plot. By using unit cell software and inputting the acquired h k l values of several intensity peaks in figure 2, it is evident that the peaks exhibit similarities in both pure and dye-doped materials. These values also closely correspond to the reported values in table 2. No new peaks were observed and no variations in peak were detected. The diffracted intensity peaks of the dye-doped material are smaller compared to the pure specimen due to the absorption of the dye in the L-threonine matrix. Introducing methylene blue into L-threonine crystals result in decreasing in intensity of the of the peaks exhibited in powder x-ray diffraction (XRD) patterns. Moreover, Methylene dye molecules can be familiarized into the L-threonine crystal lattice to disturb the usual arrangement of L-threonine molecules. This disturbance could affect the crystal's long-range order and periodicity, which would diminish the intensity of the diffraction peaks.

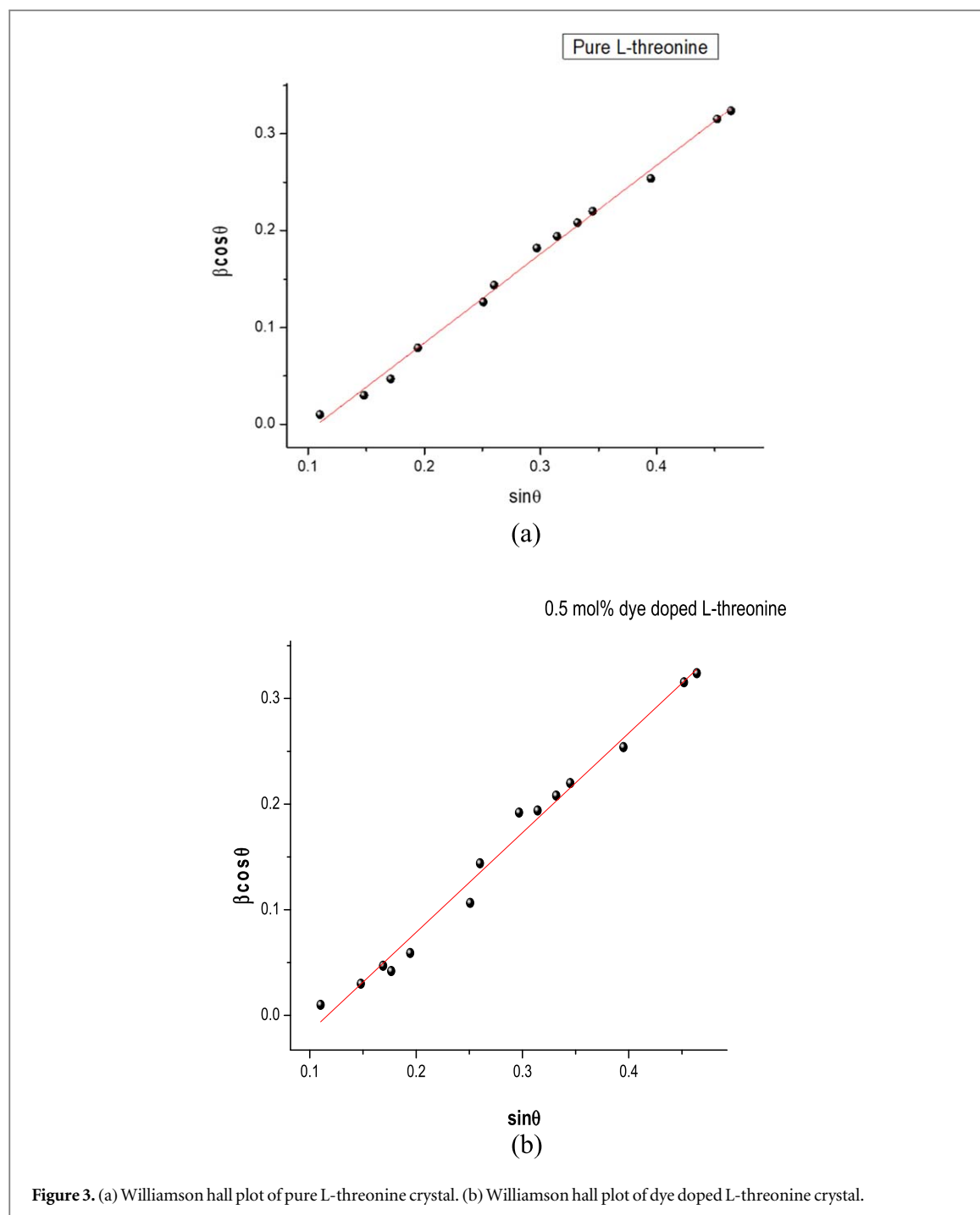
The reduction in powder x-ray diffraction (XRD) peak strength seen by introducing methylene blue into L-threonine crystals can be ascribed to many factors:

The addition of methylene blue can result in disorder and distortion in the crystal structure of L-threonine. This disorder can cause a disturbance in the normal atomic arrangement inside the crystal, resulting in a reduction in the strength of XRD peaks. Disorder can result in the expansion and attenuation of diffraction peaks, leading to a decrease in their intensity and making them more difficult to differentiate. Doping can also cause the introduction of flaws and irregularities in the crystal lattice, including vacancies, interstitials, or substitutional defects. These imperfections have a higher ability to disperse x-rays compared to a flawless arrangement of atoms, resulting in a decrease in the highest level of intensity due to greater scattering and less coherence. The presence of methylene blue molecules at lattice sites can disrupt the diffraction pattern by causing x-rays to scatter, resulting in a decrease in the strength of diffraction peaks. The interference arises from the distinct scattering characteristics of the dopant molecules in comparison to the host crystal lattice.

Methylene blue has the ability to cause specific alterations in the nearby structure of dopant molecules, which in turn impacts the arrangement of atoms and the overall symmetry of the crystal. Local structural changes have the potential to modify the XRD pattern, resulting in reduced peak intensities. Doping has the ability to affect the dimensions and structure of crystallites present in the powder sample. Variations in the size or morphology of crystallites can impact the diffraction pattern, especially the intensity of peaks, because of the distinct scattering properties of crystallites of varying sizes or shapes.

3.3. Williams hall plot

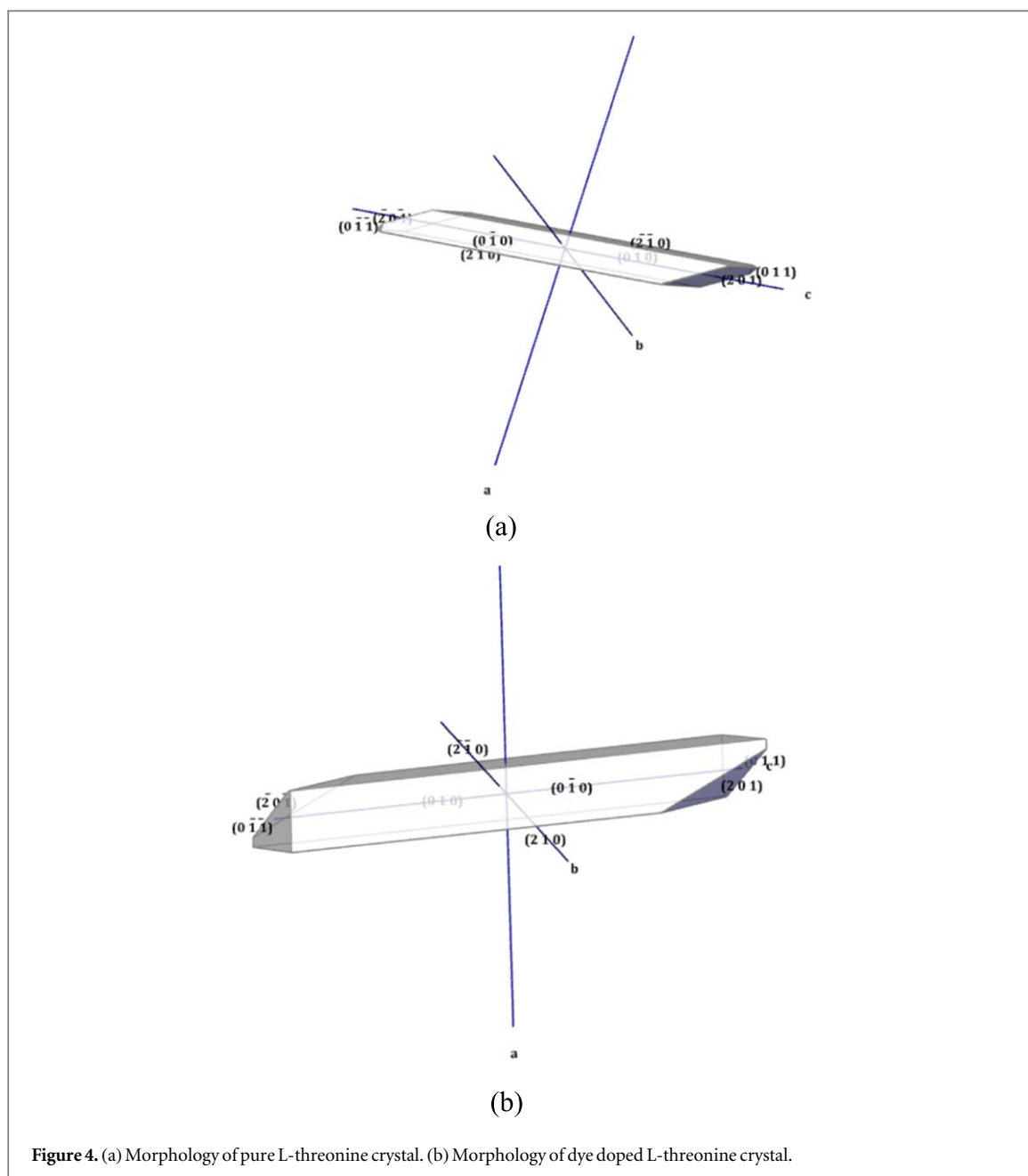
Williamson and Hall introduced a technique to separate the effects of size and strain broadening by analysing the variation in peak width with respect to 2θ . Based on this assumption, a mathematical connection was developed between integral breadth (β), volume weighted average size (D), and micro strain (ϵ) [15].



$$\beta \cos \theta = \frac{k\lambda}{D} + 4 \sin \theta$$

Figures 3(a) and (b) show the gradients of the linear regression analysis conducted on the WH plot for both pure and doped crystals, respectively. The strain value of the linear regression for the lattice with dye incorporation rises from 0.05144 to 0.07761, indicating a small change in lattice strain. The slope of the linear regression on the WH plot provides the intrinsic stress value, while the Y-intercept provides the particle size value determined for both the pure matrix (2.7037×10^{-10} m) and the matrix with dye (1.18×10^{-8} m). The nucleation and development of L-threonine crystals could be influenced by the presence of methylene blue molecules during the process of crystallisation. This impact can lead to the formation of bigger crystals with an augmented particle size in comparison to crystals that are not doped. Methylene blue molecules can engage with L-threonine molecules by many mechanisms, including as hydrogen bonding, van der Waals forces, and electrostatic interactions.

These interactions can facilitate the creation of bigger crystal aggregates or clusters, resulting in an augmentation in particle size. The addition of methylene blue can affect the Ostwald ripening process, which is



the phenomenon in which bigger crystals develop by consuming smaller ones. This can lead to a general rise in particle size as bigger crystals become more prevalent in the crystal population. The particle size of the doped crystals can be influenced by several factors, including the circumstances under which the crystals are produced, such as temperature, solvent composition, and concentration of dopant molecules. The inclusion of methylene blue under optimal circumstances can promote the development of bigger particles during crystal growth. The process of doping methylene blue can result in the agglomeration of L-threonine crystals, wherein individual crystals coalesce to create bigger particles. The process of agglomeration can lead to an augmentation in the size of particles.

3.4. Morphology analysis of pure and dye incorporated l-threonine crystal

The manufacturing of devices is a critical factor that relies on the proper shape of a crystal produced in NLO and optoelectronic devices. The growth rate of crystals varies due to the anisotropic nature of crystalline samples in various plane orientations. The morphology of both the pure crystal and the crystal with dye incorporation is shown in figures 4(a) and (b) respectively, using the winmorph programme. Sharp peaks were seen in the PXRD investigation. The crystal's thickness rose uniformly across all planes due to the infiltration of dye admixture atoms. The dislocation occurring in the crystal enhances the growth rate along the *c*-axis. The prominent faces are (2 1 0), (0 $\bar{1}$ 0), (0 1 1), (2 0 1) and existence of corresponding Friedel planes in pure and doped crystals.

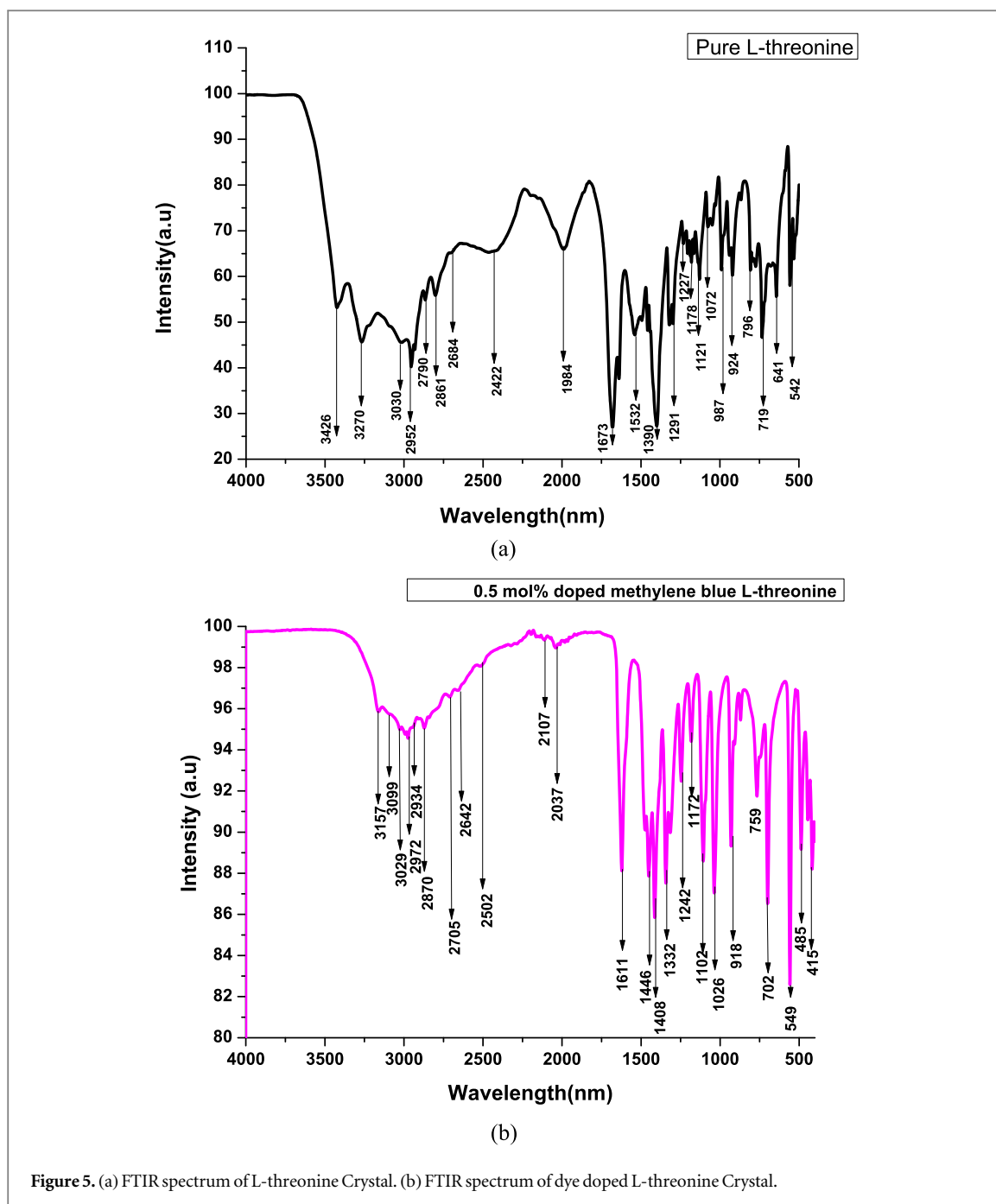


Figure 5. (a) FTIR spectrum of L-threonine Crystal. (b) FTIR spectrum of dye doped L-threonine Crystal.

3.5. FTIR analysis

A spectroscopic analysis is conducted using FTIR spectroscopy to examine different functional groups in a clean matrix and a matrix doped with dye. The observed fingerprint area provides information about the molecular properties of both the pure matrix and the matrix doped with dye. The frequencies of the incidents are observed within the range of $400\text{--}4000\text{cm}^{-1}$, as shown in figures 5(a) and (b). There are slight differences in the location of the highest point of the dye-infused matrix compared to the pure matrix. The appearance of additional peaks at 2972cm^{-1} , which is attributed to the methylene group, and 549cm^{-1} shows that there is a dye admixture present in the pure matrix. Moreover, the peak strength falls in the dye admixture sample compared to the pure sample due to the impact of the blue dye colour on the pure L-threonine sample, which is originally white in colour. The band assignments are stated in table 3. The acquired values are consistent with the findings reported in previous research [11].

3.6. UV–vis analysis

The material's linear optical behaviour was studied by putting a crystal sample with a mirror-like surface to a UV–vis spectrometer. The crystal sample consisted of a pristine matrix and a matrix doped with dye.

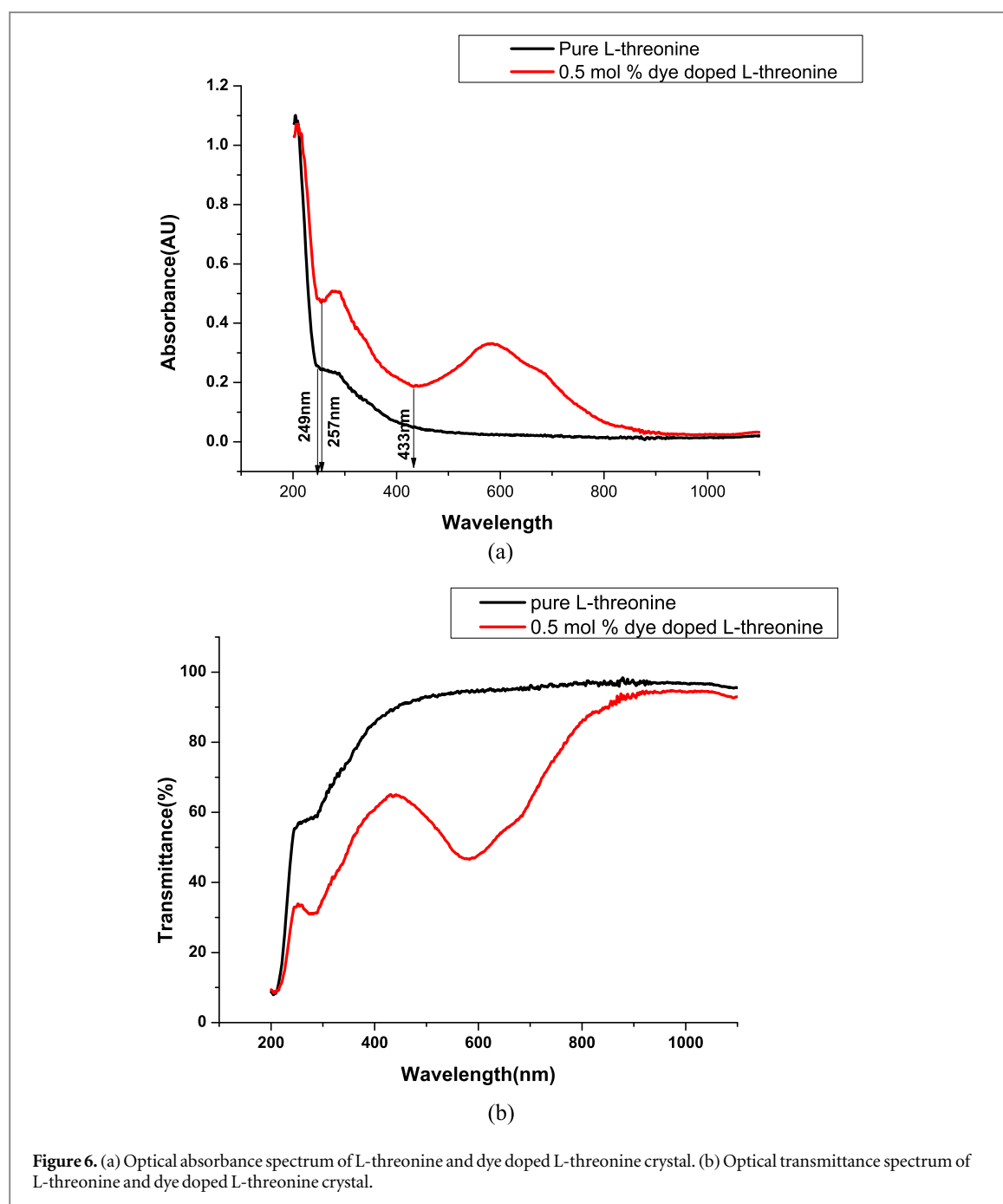


Figure 6(a) displayed the absorption spectra. It has been noted that the absorption is enhanced in the higher wavelength range at 257 nm when compared to the pure L-threonine crystal. The observed peak at 433 nm is a result of the presence of methylene blue molecules mixed with a pure L-threonine matrix. The presence of the CH_3 chromophore in the structure is responsible for the blue colour of the dye when mixed with pure L-threonine, which has a transmittance of 90% [16]. This minimises the breadth of the optical zone illustrated in figure 6(b). However, the dopant lattice is more important for applications that need selected wavelengths of light [17]. The inclusion of dye in the lattice decreases the number of scatterings centres and enhances the absorption of wavelengths throughout the matrix. This results in an increase in bandgap compared to the original matrix, making it more suitable for nonlinear optical (NLO) applications.

A broader bandgap is often preferred for nonlinear optical (NLO) applications for the following basic reasons: transparency across a wider wavelength range is ensured by a greater bandgap, especially in the visible and near-infrared ranges. This has significance for NLO materials, since light absorption at operating wavelengths might lower the efficiency of the nonlinear optical processes. Narrower bandgap materials tend to absorb photons more readily in NLO applications, which leads to higher losses and worse efficiency. Because a larger bandgap lowers these losses, more light can pass through the material without being absorbed. In NLO

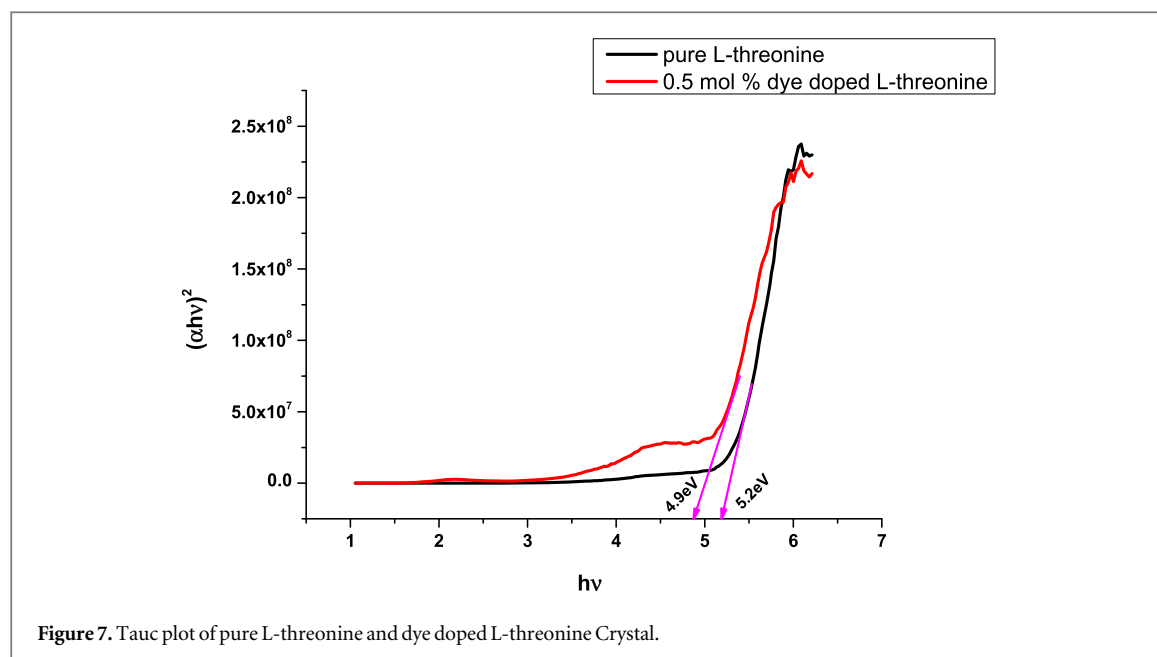


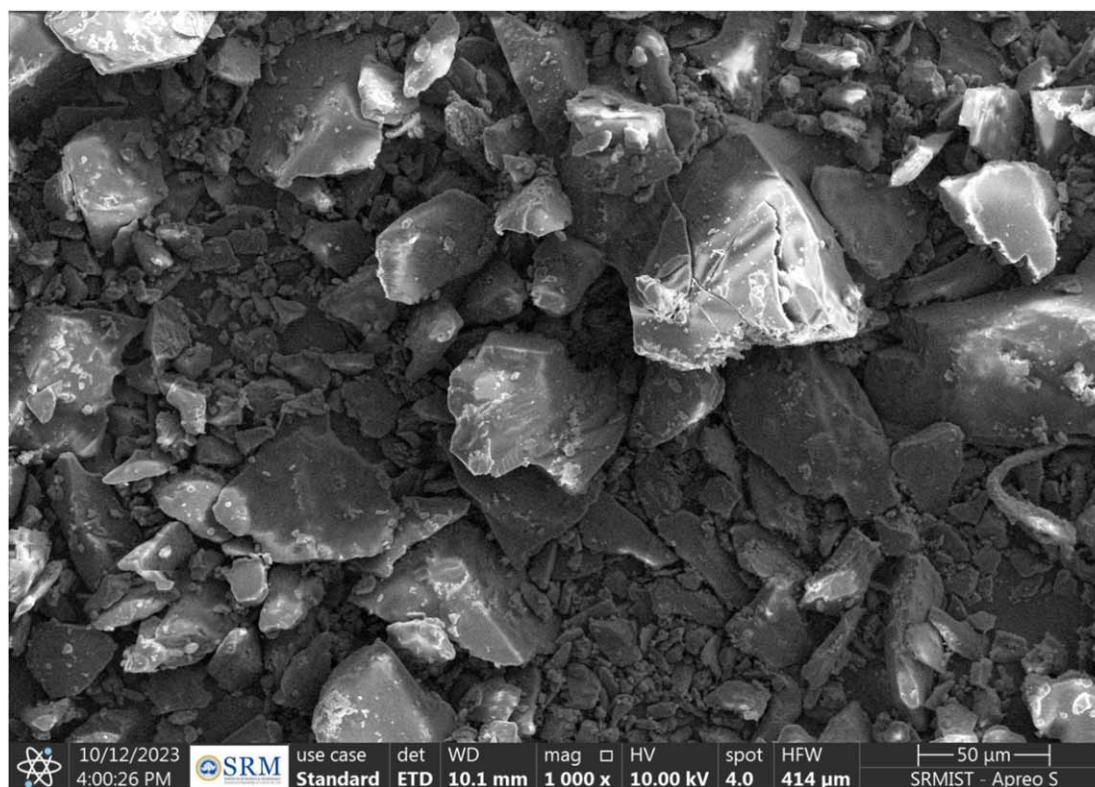
Figure 7. Tauc plot of pure L-threonine and dye doped L-threonine Crystal.

Table 3. Vibrational Band Assignments for dye doped L-threonine Crystal.

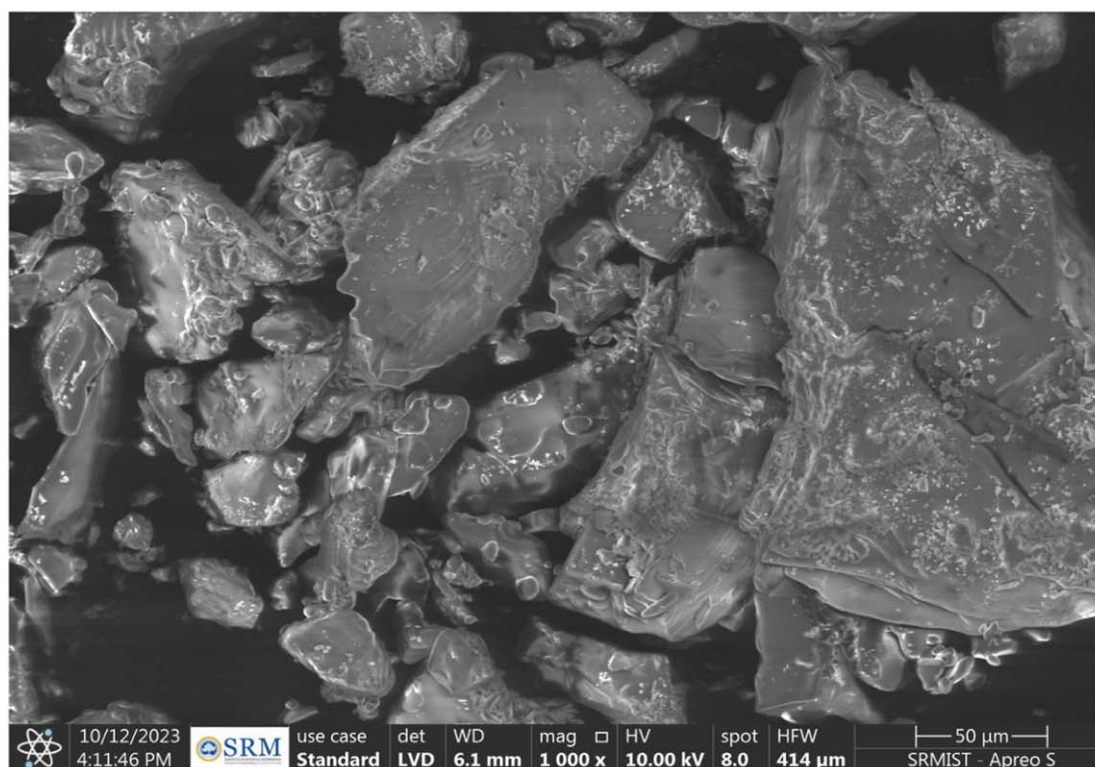
Pure l-threonine	Dye doped l-threonine	Assignment mode
3426	—	OH stretching vibrations
3270–3030	3157–3099	Asymmetric stretching vibration of NH_3^+
2952	3029	Symmetric Stretching vibrations of NH_3^+
—	2972–2934	Stretching vibrations of dimethylamino group $(\text{CH}_3)_2\text{NH}$ of methylene blue molecules [19]
2861	2870	Superimposed OH and NH_3 stretching vibrations
2790	2705	
2684	2642	
2422	2502	
	2107	
1984	2036	
1673	1611	C = N stretching/C = C stretching
1532	1446	COO^- asymmetric stretching
1390	1408	COO^- symmetric stretching
1291	1332	Bending vibrations of CH
1121	1102	Rocking of NH_3^+
1072	1026	C-N stretching of amino group
924	918	C-C stretching
796	759	Torsion of COH
719	702	Wagging vibration of CO_2^-
—	549	C-S-C stretching of Methylene blue molecules [19]
542	485–415	Torsional mode of NH_3

operations, high-intensity light is often used, which might lead to two-photon absorption if the bandgap is too small. Two-photon absorption is one nonlinear absorption mechanism that might mitigate and reduce the impact of the desired nonlinear optical effects.

Larger bandgap materials often exhibit improved thermal stability. High-intensity laser light can cause deterioration or modify the properties of materials with smaller bandgaps. Greater heat load tolerance can be shown by materials having a greater bandgap. Though not directly related to bandgap size, larger bandgap materials can sometimes have more advantageous electrical topologies for nonlinear optical phenomena. This can give rise to higher second- or third-order nonlinear coefficients, which are required for efficient frequency conversion and other NLO operations. Larger bandgap materials are often associated with higher optical damage thresholds. This allows them to withstand higher laser light intensities without breaking, making them more robust and trustworthy for high-power NLO applications. The combination of these characteristics renders larger bandgap materials more advantageous in several nonlinear optical applications, including but not limited to optical parametric amplification, frequency doubling (second harmonic synthesis), and associated procedures.



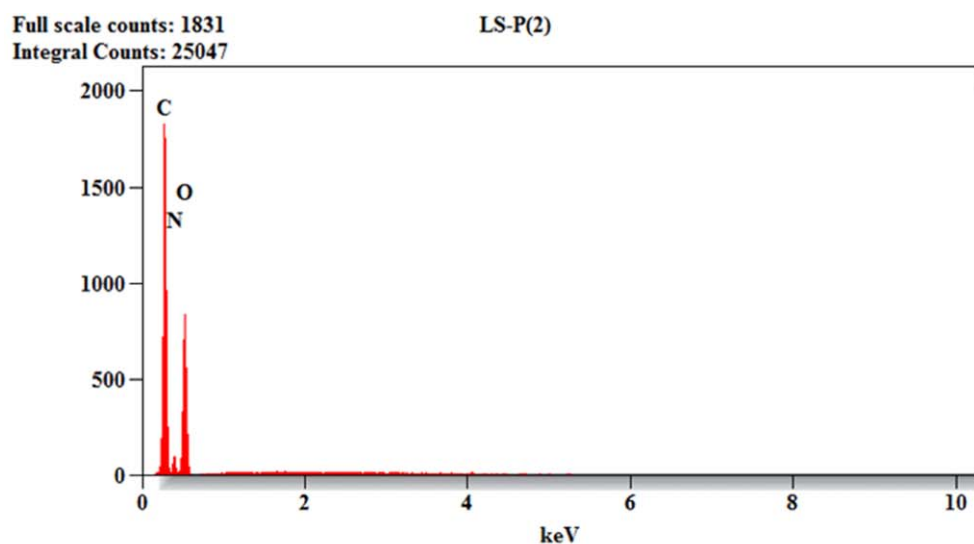
(a)



(b)

Figure 8. (a) SEM image of pure L-threonine Crystal. (b) SEM image of dye doped L-threonine Crystal.

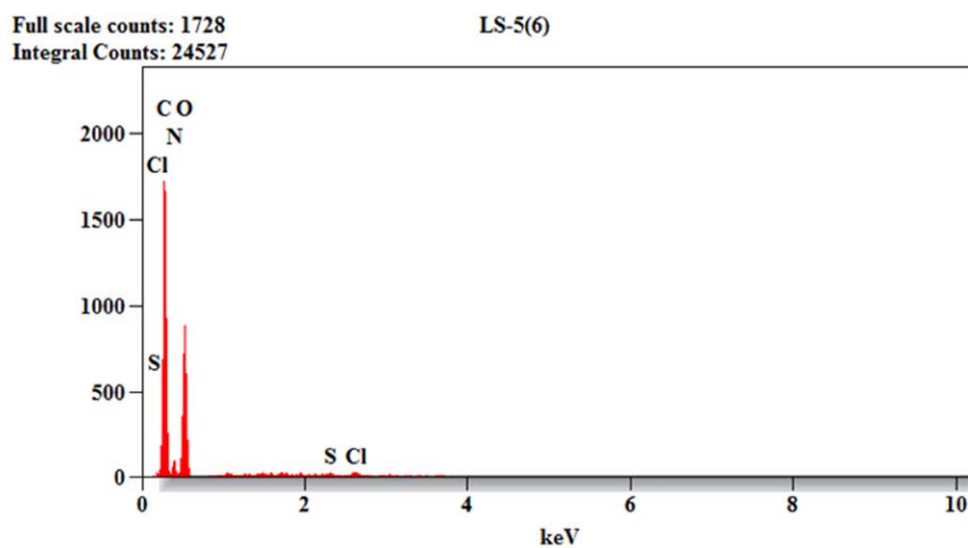
The transfer of electrons from the conduction band to the valence band in high band gap materials affects the expulsion of direct band gap. The direct band gap values for both the dye-incorporated and clean samples are shown. At an energy level of 5.2 electron volts (eV), the photon absorption coefficient is shown in figure 7. The presence of methylene blue dye molecules decreases the absorbance coefficient at 4.9 electron volts (eV), resulting in noticeable changes [18, 19].



Quantitative Results for: LS-P(2)

Element	Net Counts	Weight %	Atom %	Atom % Error	Formula
C	9956	29.69	35.21	± 0.35	C
N	1026	17.19	17.48	± 1.55	N
O	5068	53.13	47.31	± 0.84	O
Total		100.00	100.00		

(a)

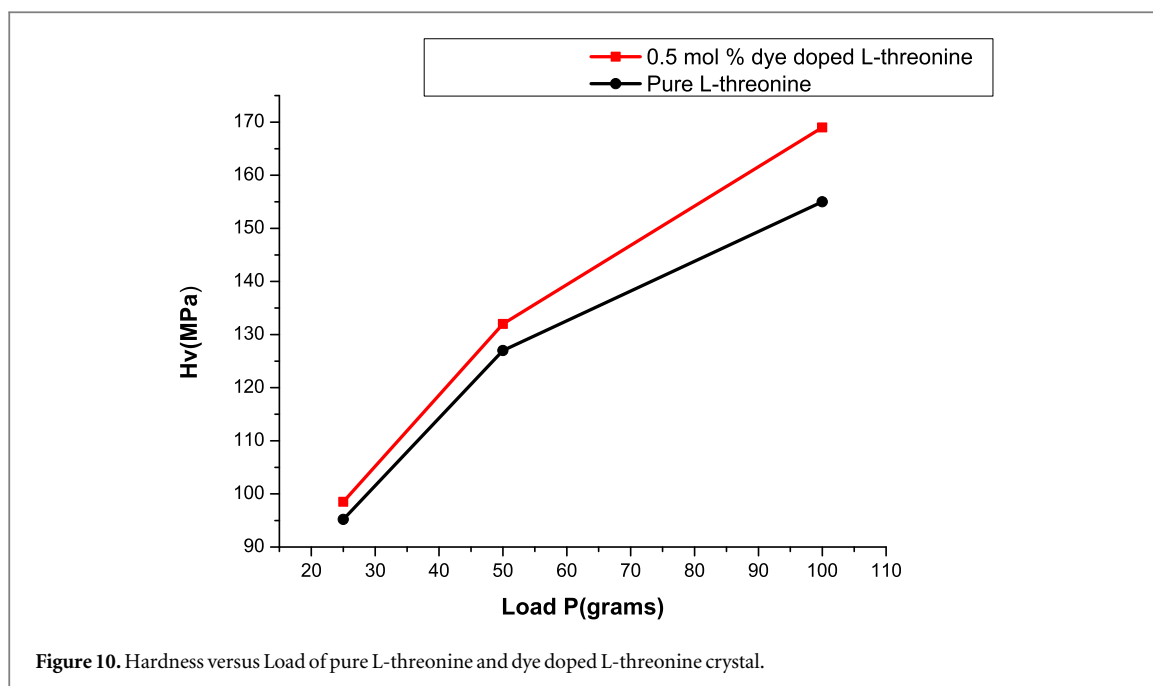


Quantitative Results for: LS-5(6)

Element	Net Counts	Weight %	Atom %	Atom % Error	Formula
C	9353	29.80	35.46	± 0.37	C
N	891	15.61	15.92	± 1.64	N
O	5231	54.29	48.49	± 0.86	O
S	72	0.09	0.04	± 0.02	S
Cl	140	0.21	0.08	± 0.02	Cl
Total		100.00	100.00		

(b)

Figure 9. (a) ETAX spectrum of pure L-threonine Crystal. (b) ETAX spectrum of dye doped L-threonine Crystal.



Specific electronic transitions between the dye molecules in methylene blue and the host L-threonine crystal can be the cause of the peak at 4.5 eV. These changes can result from excited electrons to higher energy states, as shown by peaks in the absorption spectra. Impurity states could have been introduced into the band structure of the L-threonine crystal using methylene blue. The possibility exists that these states facilitate electronic transitions at energies different from the host crystal's intrinsic bandgap, leading to additional peaks in the absorption spectrum. Methylene blue doping can result in the formation of charge transfer complexes between the dye molecules and the host crystal. These chemicals have distinct absorption properties, which might be responsible for the peak at 4.5 eV. Methylene blue molecules have the potential to generate localised electronic states inside the bandgap of L-threonine. On the tau plot, transitions involving these localised states might show up as additional peaks. Excitons, or bound electron-hole pairs, have unique absorbance characteristics and can arise from the interaction of methylene blue with the L-threonine crystal lattice. This might result in peaks about 4.5 eV, for example.

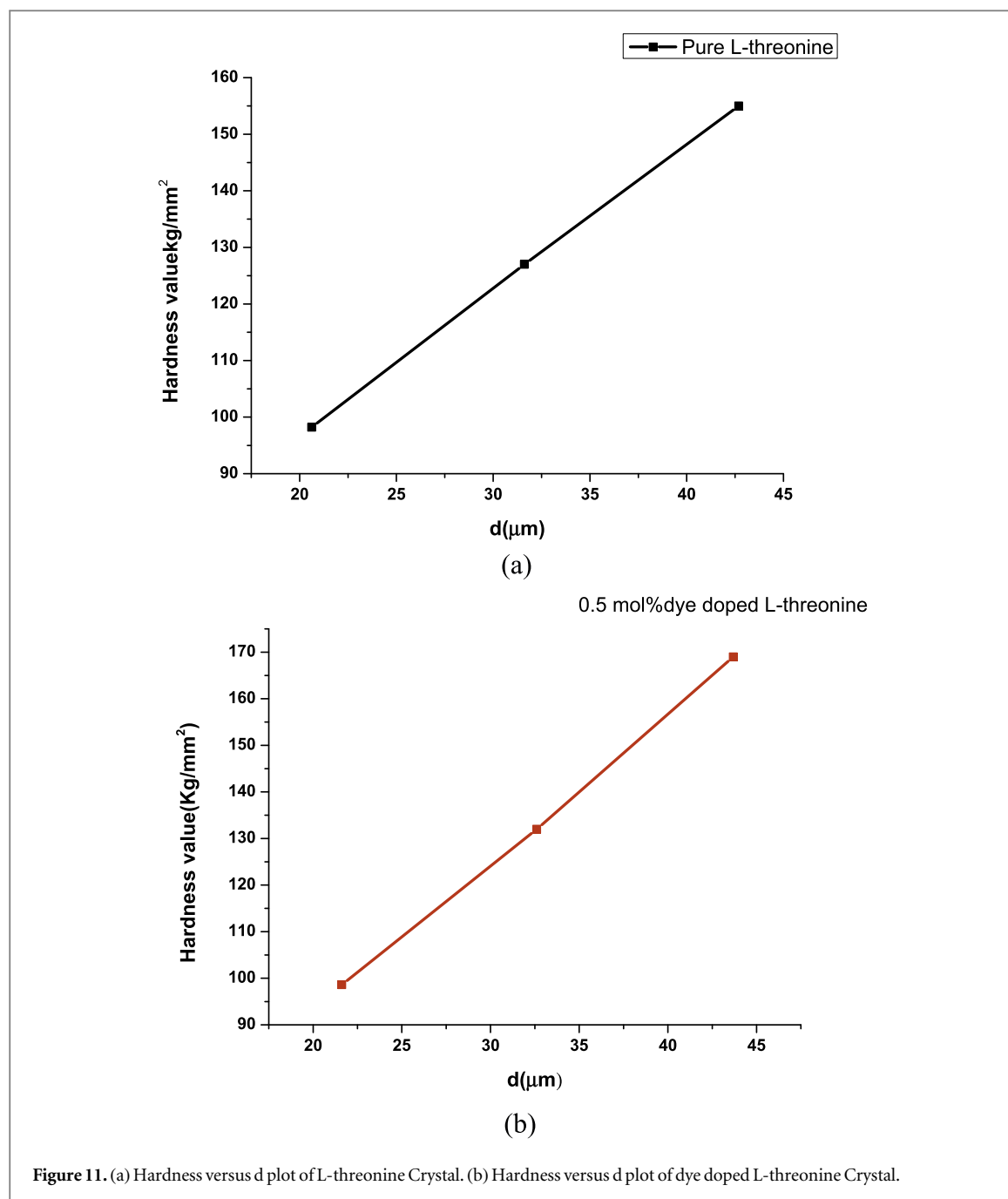
3.7. SEM-EDAX analysis

The scanning electron microscope (SEM) images of the pristine sample and the pristine sample with included dye are shown in figures 8(a) and (b), respectively. An electron beam investigation conducted in a pattern of parallel lines is used to analyse surface imperfections in a formed crystal. This examination reveals that certain locations on the sample supply the necessary information about the specimen. The presence of dye inclusions on the crystal surface during the crystallisation process leads to an increase in the size of the doped crystal compared to the pure crystal. Additionally, it also results in an increase in surface dislocations. Therefore, it modifies the result derived from linear optical investigations.

The EDAX characterization assists in confirming the presence of dye components embedded in the clean matrix. Figures 9(a) and (b) show the quantification of elements in both the pure matrix and the matrix with dye incorporation. No sulphur or chlorine was detected in the pure L-threonine amino acid. The presence of sulphur and chlorine molecules in the dye, together with its distinct peak, demonstrates that methylene blue dye has been integrated into a pure L-threonine amino acid matrix.

3.8. Hardness analysis

The mechanical hardness value of a crystal is a measure of its qualitative strength and resistance, specifically for single crystals. In dye admixture single crystals, the non-linear coefficient is often reduced to a lower value in the damaged and defective segments compared to the pure L-threonine single crystal. Exceptional grade crystals must possess outstanding optical non-linear performance as well as impressive mechanical properties. The Futuretech FM-800 E series underwent Vickers hardness testing to evaluate its mechanical qualities, including yield strength, cracking temperature, and splinter deflection [20].



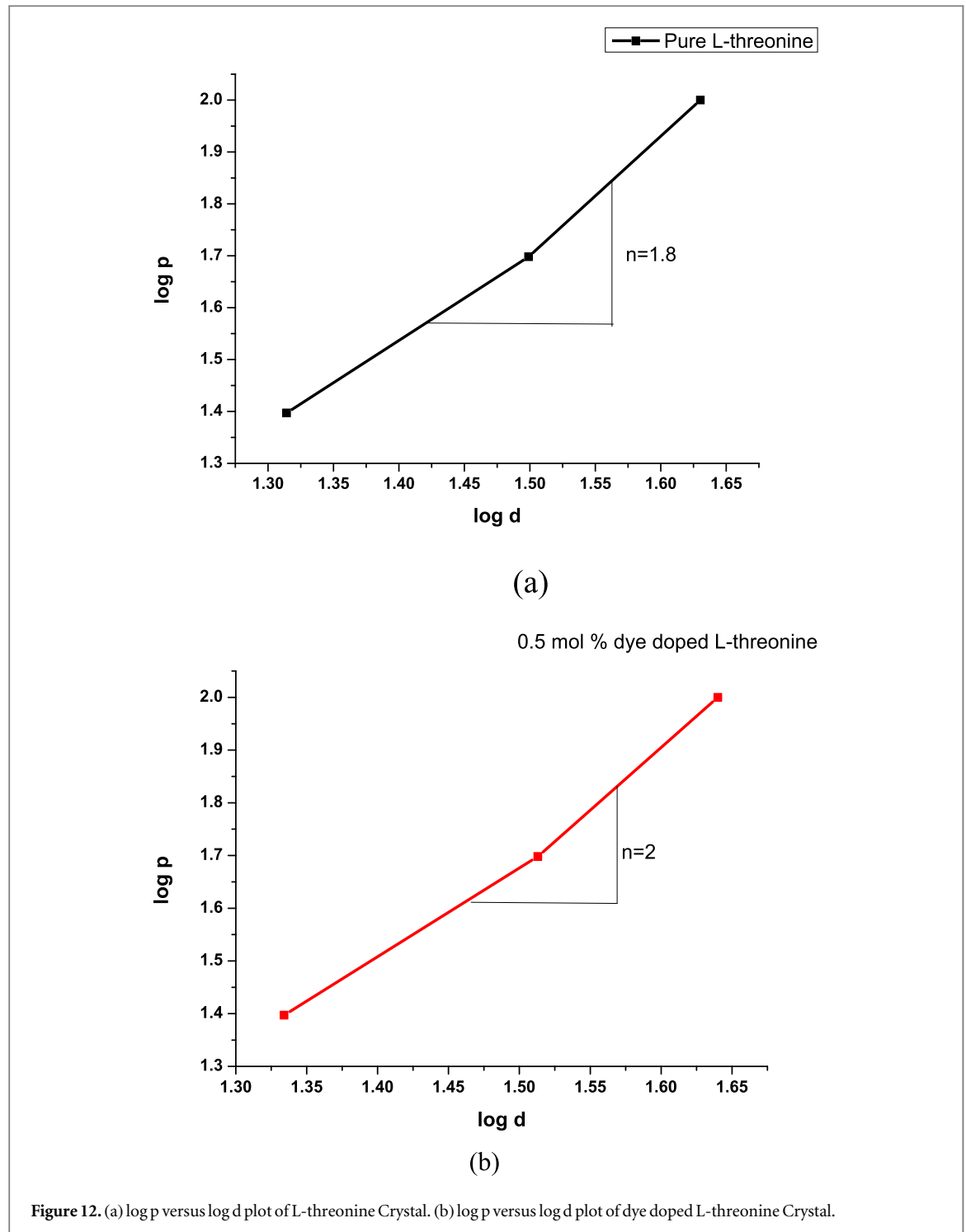
The Vickers diamond pyramidal number, H_v , was obtained using the following equation:

$$H_v = \left[\frac{1.8544}{d^2} \right] \text{Kg mm}^{-2} \quad (1)$$

Where P represents the weight applied in kilogrammes and d is the diagonal length of the depression in millimetres.

Increased hardness is achieved when a crystal is fully formed, requiring significant stress to cause dislocations in the crystalline matrix, resulting in enhanced crystalline fineness. The increased hardness values shown in figure 10 for dye-doped L-threonine can be attributed to the presence of methylene blue ions mixed with the L-threonine matrix. In order to create a linear chain, molecules having symmetry in their respective planes are connected to each other using identical connections.

$P = ad^n$, where the functional load P and diagonal length d of the indenter are obtained by substituting 'a' for an arbitrary constant, 'n' for the work hardening coefficient for a particular material, 'd' for the Indentation diagonals Impression in milli meter, and 'P' for the Load in gram depicted in figures 11(a) and (b). Caner's conditions are satisfied by the almost perfectly straight log P versus log d figure in 12(a) and figure 12(b). For both hard and soft material, Onitsch statement work hardening coefficient n lies between 1.0 to 1.6 for hard



materials and greater than 1.6 are soft materials [21]. By using the least-squares fit approach, it is determined that a completely developed crystal exhibits a work hardening coefficient of 2 at standard ambient temperature.

The single crystal is hence relatively malleable. By using [22], one can ascertain the yield strength for values of n that are less than 2 based on hardness.

$$\sigma_y = \frac{H_v}{3} \quad (2)$$

The stiffness of the sample is denoted by H_v , whereas the yield potency (hardness) is represented by σ_y . Figures 13(a) and (b) illustrate the yield potency (σ_y) of dye-doped crystal specimens at normal ambient temperature.

The adhesive force between a clean matrix and a matrix doped with dye is determined by the stiffness constant. The free energy of a dye-incorporated single crystal experiences a significant rise before it breaks. The sample's elastic stiffness constant (C_{11}) is computed at standard temperatures using Wooster's [23] empirical

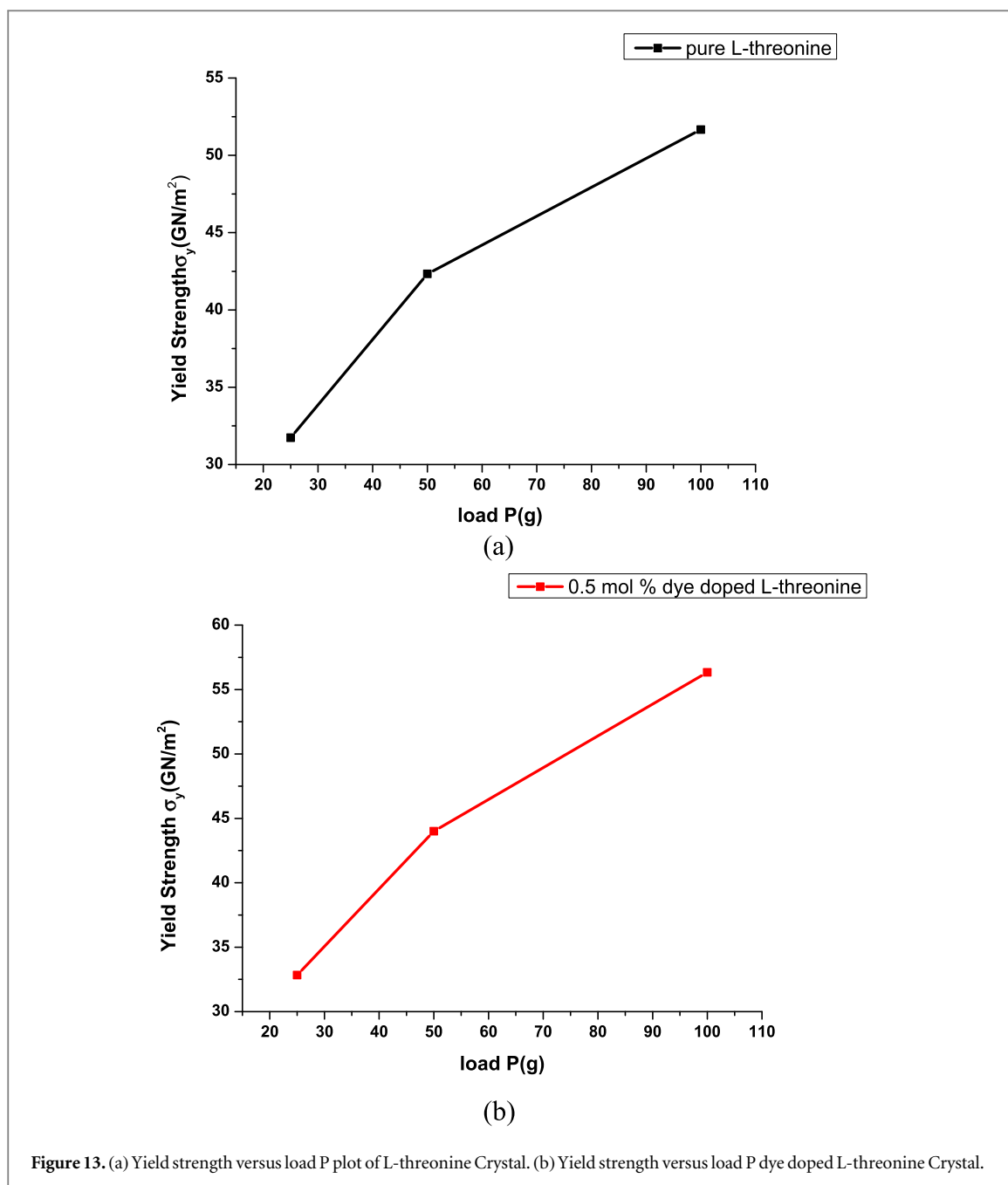


Figure 13. (a) Yield strength versus load P plot of L-threonine Crystal. (b) Yield strength versus load P dye doped L-threonine Crystal.

formula.

$$C_{11} = H_V \frac{Z}{4} \quad (3)$$

The stiffness constant graph, seen in figures 14(a) and (b), is similar to the hardness integer graph. It provides a representation of the bonding tightness in the pure and dye doped L-threonine matrix. Table 4 displays the measurements of Vickers hardness (H_V), yield strength (σ_y), and stiffness constant (C_{11}) for both pure crystals and crystals with dye admixture. The grades mentioned above demonstrate the correlation between mechanical stability and crystallinity. The link between the number of dislocations in completely formed crystals and microhardness was established, showing that fewer dislocations resulted in higher microhardness. This material seems to be very efficient for use in NLO modulators, band pass filters, and wavelength selective photonic applications [24]. It has a high yield potency, hardness value, and resistance to damage caused by lasers.

3.9. DTA/TG analysis

Thermogravimetric/differential thermal analysis (TG/DTA) was conducted on both pure L-threonine and a combination of L-threonine single crystal with dye. The resulting data were graphed in figures 15(a) and (b). This research is a crucial instrument for determining the endothermic peak and decomposition temperature of a

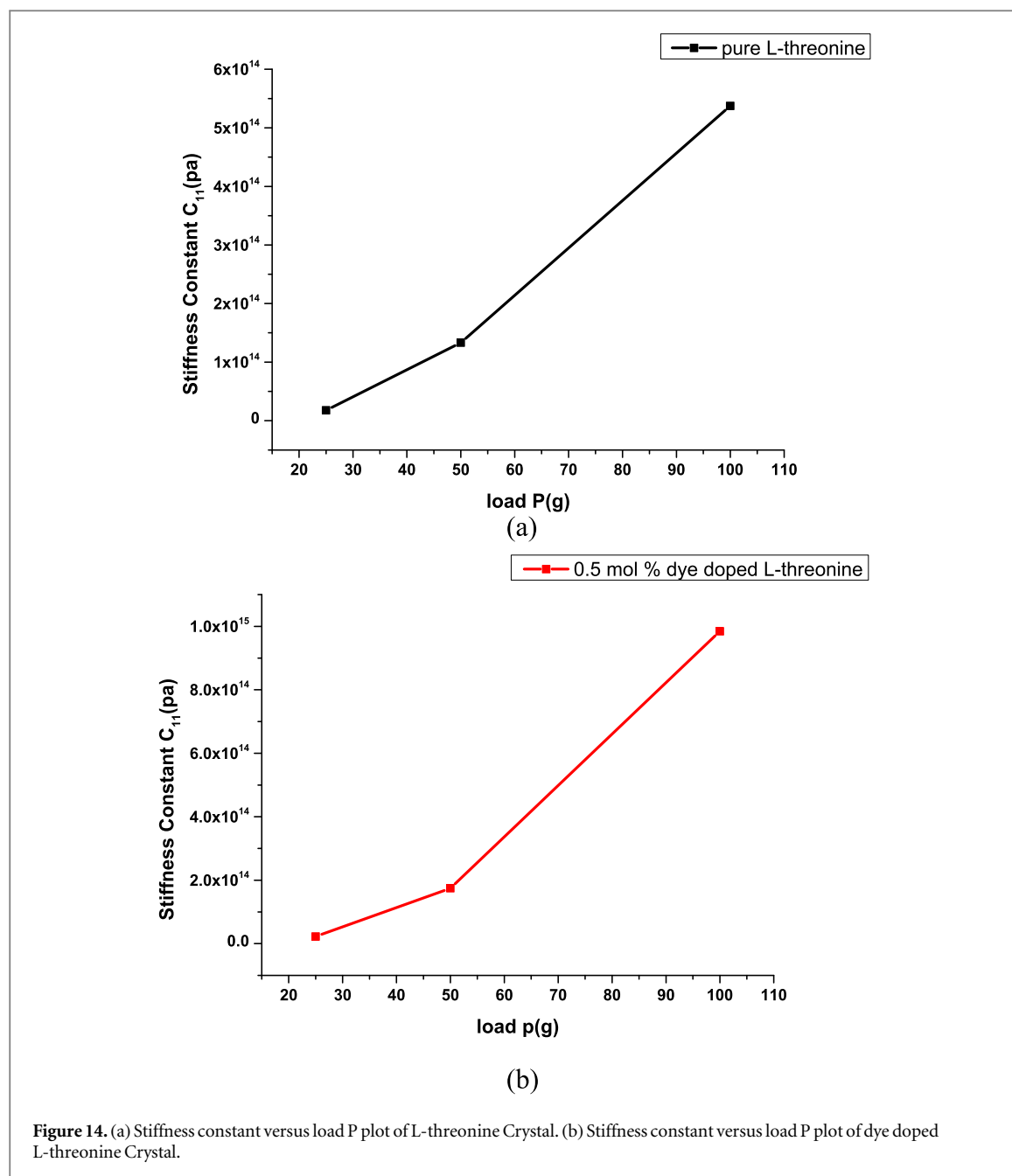
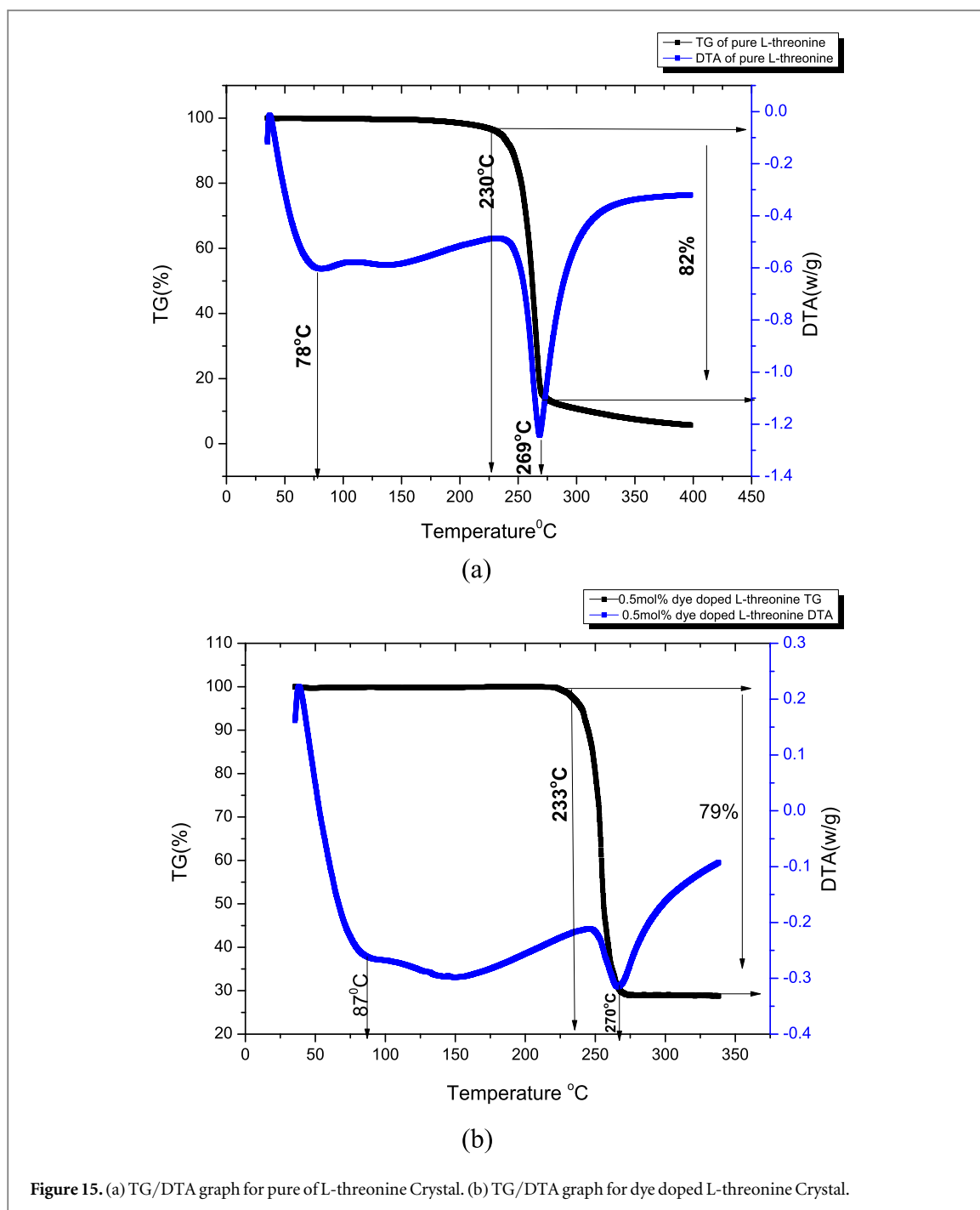


Table 4. Microhardness (H_v), yield strength (σ_y), stiffness constant (C_{11}), fracture toughness of Pure and dye incorporated L-threonine crystal.

Load P (g)	H_v (kg mm ⁻²)		σ_y (GN m ⁻²)		$C_{11} \times 10^{14}$ (Pa)	
25	95.2	98.5	31.7333	32.8333	0.177	0.225
50	127	132	42.33333	44	1.33	1.75
100	155	169	51.66667	56.33333	5.37	9.84

material. The graph clearly indicates that the dye admixture crystal has a higher disintegration temperature than the pristine crystal. This is due to the presence of dye molecules within the matrix of the pure amino acid l-threonine crystal. The decomposition temperature of the material begins at 230 °C for Pure & 233 °C for doped L-threonine which reveals the presence of doped methylene blue molecules (0.5 mol%). This causes the breaking of bonds and the burning of volatile compounds. Additionally, the material experiences a single weight loss of 82% starts at 230 °C for Pure and 79% starts at 233 °C for doped L-threonine of its initial mass, indicating complete saturation. The pure and dye doped material has a melting point of 269 °C & 270 °C which have been shown by a distinct endothermic peak, suggesting that the maximum temperature corresponds to a phase transition.



Thermal decomposition of L-threonine occurs at high temperatures, resulting in the breakdown of smaller molecules into gases (CO_2 , NH_3 , H_2O) and other volatile chemicals. As a consequence of these smaller molecules evaporating, there is a noticeable weight decrease. In particular, the carboxyl group (COOH) and amine group (NH_3) are prone to heat breakdown that results in gas release. Additionally, L-threonine can remove water molecules from the body, which helps people lose weight in hot environments. This process, called dehydration, usually happens before the amino acid undergoes total thermal breakdown. The thermal breakdown of materials in an inert environment at high temperatures is known as pyrolysis. Pyrolysis of L-threonine can produce gaseous products and char, which can significantly reduce its mass.

The observed decrease in the single weight loss event during thermal analysis when L-threonine is doped with 0.5 mol% methylene blue dye could be due to many factors: Methylene blue doping of L-threonine efficiently reduces its concentration. The mass of L-threonine accessible for breakdown is somewhat lower since methylene blue makes up just 0.5 mol% of the sample as a whole. This might explain why the weight loss percentage observed is lower.

Table 5. SHG efficiency of dye doped L-threonine crystal.

Crystal	Output in mJ	Comparing the second harmonic generation (SHG) efficiencies with respect to potassium dihydrogen phosphate (KDP) with an efficiency of 1.0 scaling.
KDP	7.5	1
Dye doped L-threonine	16.5	2.2
Pure L-threonine	8.6	1.15

L-threonine and methylene blue can interact to stabilise the latter's structure and improve its thermal stability. The weight loss at the initial decomposition temperature can be decreased as a consequence of this interaction, which can raise the decomposition temperature. The breakdown route of L-threonine can change in the presence of methylene blue. This suggests that rather than resulting in a sudden, dramatic weight loss event, the breakdown can occur via a variety of intermediates or phases. In summary, a combination of stabilisation effects, modified breakdown routes, and physical interactions between the dye and the L-threonine molecules is probably responsible for the decrease in the single weight loss event for methylene blue-doped L-threonine.

3.10. SHG analysis

Q-switched The QUANTA RAY Model LAB — 170 — 10 is a powerful laser that uses Nd:YAG technology. In crystals exhibiting second harmonic generation (SHG), each pulse requires two millijoules of input energy. The incident light has a wavelength of 1064 nm, whereas the emitted light from the sample has a wavelength of 532 nm. The frequency of repetition is 10 Hz. The duration of the pulse is 6 nanoseconds. Characterisation approaches provide insights into the second order efficiency of a material. The virgin and dye-doped L-threonine sample was finely ground and filtered via an appropriate sieve. It was then exposed to a Nd:YAG laser with a wavelength of 1064nm. The efficiency of the Second order Harmonic Generation (SHG) for the dye admixture L-threonine was higher compared to pure L-threonine, as seen in table 5. According to the findings, the dye doped L-threonine single crystal had a conversion pulse efficiency that was 0.8 times higher than KDP [25].

3.11. Particle size dependence of SHG intensity of pure and dye doped L-threonine crystal

A screening method be used to determine whether a crystalline material with non-centrosymmetric characteristics can achieve phase matching. The materials with noncentrosymmetric crystal structures that have phase matching potential can be found by combining them with a screening technique. SHG constantly rises and approaches saturation as particle size increases, hence supporting the phase-matching features of the material. The SHG intensity's addiction to particle size was designed to promote the phase matching feature. The degree to which the SHG competence of these materials depends on the particle extent is shown [26]. Moreover, Phase matching is the matching of the phase velocities of the fundamental and second harmonic waves, and it is a prerequisite for successful SHG. Because of their short contact period, phase matching can be less significant in small particles, which might lead to less effective SHG. In larger crystals, phase matching becomes more significant and can significantly improve SHG efficiency. The coherence length is the distance over which the fundamental and second harmonic waves remain phase-locked. In larger crystals, the coherence length plays a crucial role in the efficiency of SHG. If the particle size is substantially smaller than the coherence length, phase mismatches throughout the interaction length can result in a drop in the SHG efficiency.

A sieve shaker was used to crush and sift a single pure and doped crystal into many particle sizes ranging from 65 to 125, 250 to 350, 350 to 500, and above 500 μm .

The key component of phase matching in a crystalline material is the intensification of the second harmonic generation (SHG) effect as the particle size increases, reaching saturation at the peak figure 16 illustrates the relationship between the size of crystalline minerals and their properties.

After undergoing a thorough and precise crushing and filtering procedure using several sieve shakers, the undoped and dye doped L-threonine crystals displayed particle sizes that varied from less than a micron to 500 microns. Table 6 presents the relationship between particle size and second harmonic generation (SHG). The increase in particle size is exactly proportional to the pace at which the second harmonic generation (SHG) intensity rises. The surge gradually decreases in intensity and curves around 350–5000 microns to achieve supersaturation. The relationship between particle size and SHG amplification is also seen in phase-matched crystals. This work showcases the use of crystalline materials for very effective frequency doubling and optical oscillator applications.

3.12. Temperature dependence hardness studies

Microhardness, an essential element in the manufacturing of opto-electronic devices, was used to evaluate the mechanical properties of a pure and dye-doped L-threonine crystal sample. At fixed indentation the Vickers

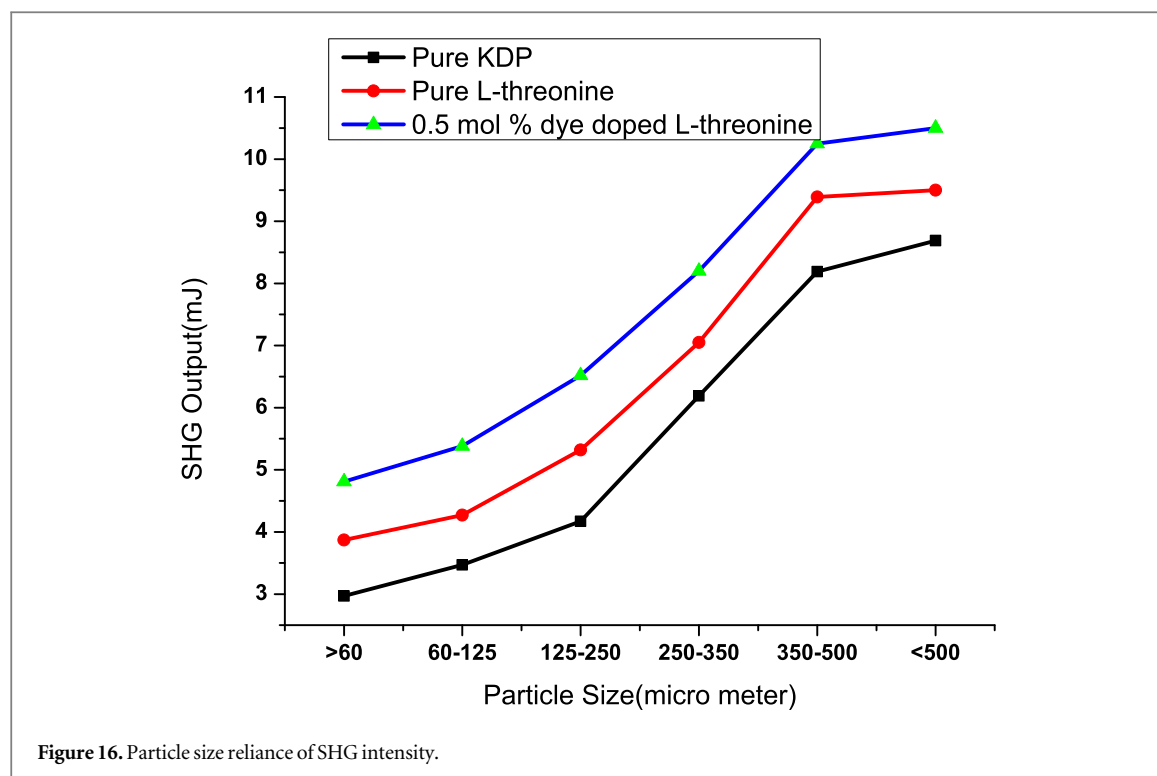


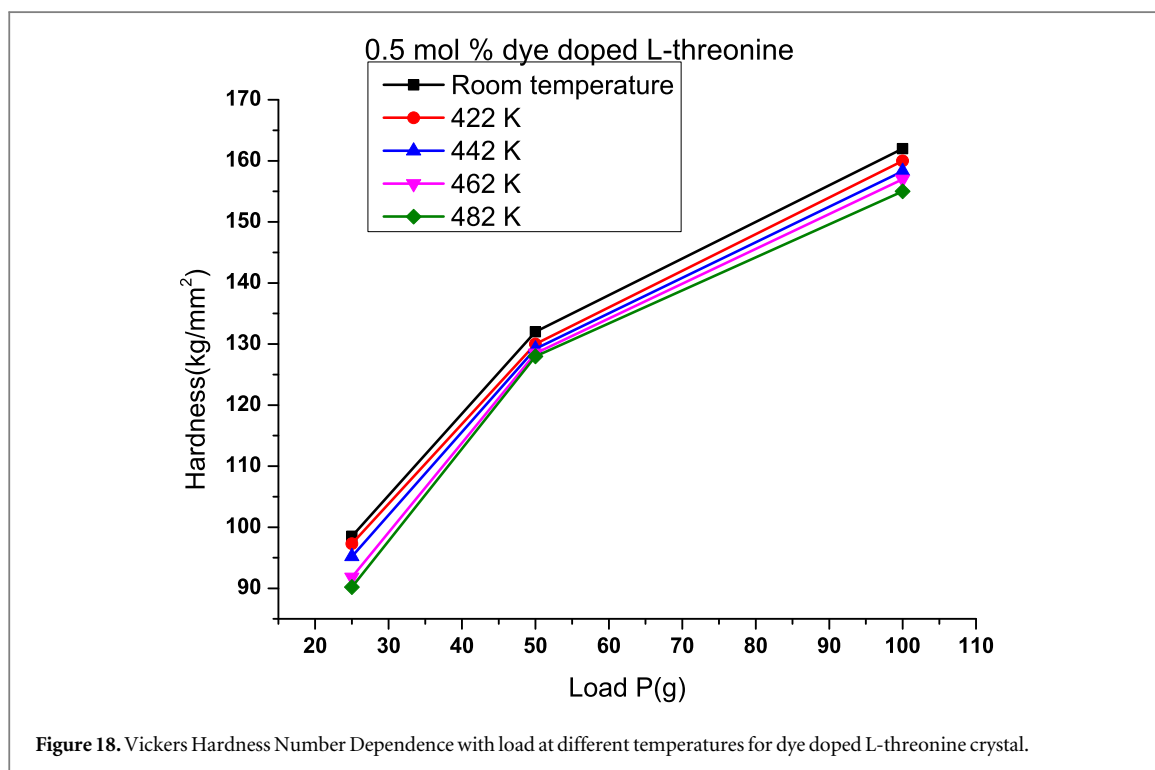
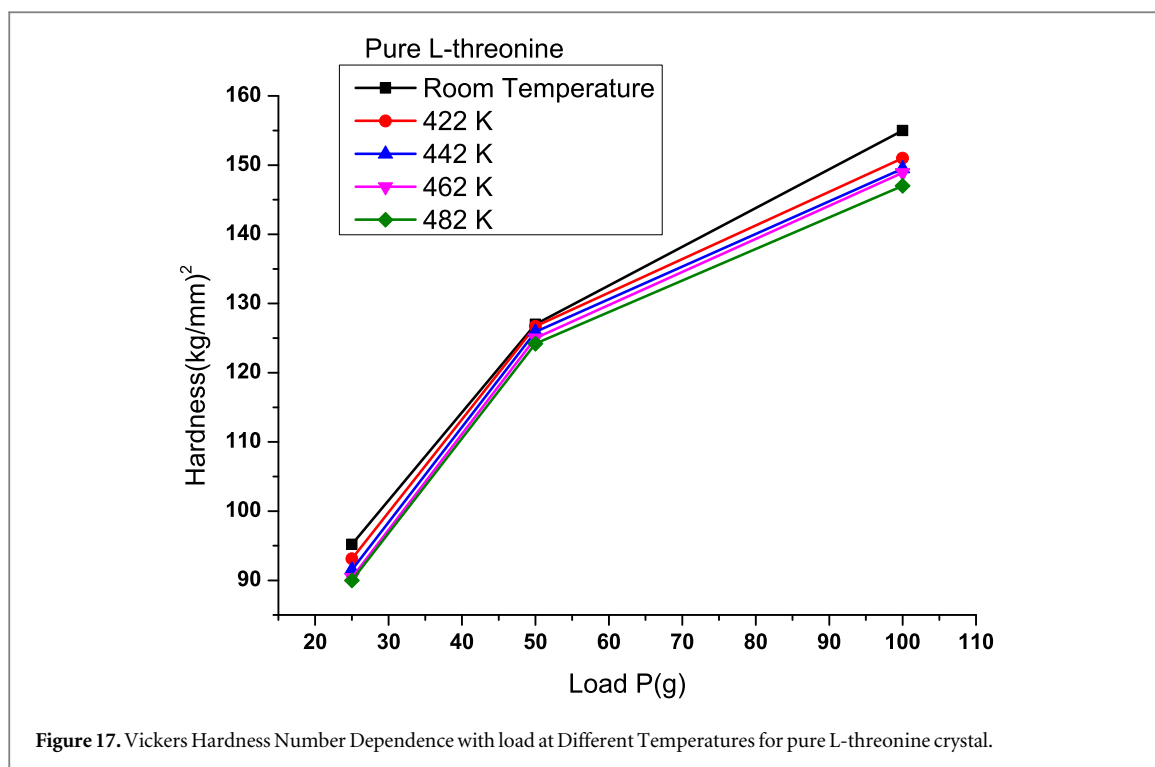
Table 6. Particle size dependence of SHG intensity of Pure and dye doped L-threonine crystal.

S no	Particle size in micro meter	SHG output in mJ for pure L-threonine	SHG output in mJ for 0.5 mol% doped L-threonine	SHG output in mJ for pure KDP
1	> 60	3.87	4.81	2.97
2	60–125	4.27	5.38	3.47
3	125–250	5.32	6.52	4.17
4	250–350	7.05	8.2	6.19
5	350–500	9.39	10.25	8.19
6	<500	8.69	10.5	8.69

microhardness test is conducted on L-threonine single crystals that are transparent and doped with dye. These crystals have flat, soft, and free of fractures surfaces. The hardness of a specimen refers to its ability to withstand plastic deformation. Indentation, bending, cutting, and scraping are procedures that result in persistent deformation. The hardness tests were conducted on undoped and 0.5 mol% dye-doped L-threonine crystals with dimensions of $5 \times 5 \times 3 \text{ mm}^3$. Microhardness data was collected from the projecting (2 1 0) face using a diamond pyramidal indenter and a Leitz-Wetzler hardness tester. The data was recorded for all applied weights (loads) over a continuous 3-s indentation period. Heat treatment temperatures ranging from ambient (303 K) to 422, 442, 462, 482, 502, and 522 K were effective for both pure and doped L-threonine single crystals. In the past, a range of loads from 25 to 100 g were applied to the sample's raised surface to produce indentations, which were then used to measure the microhardness (H_v). The crystals were subjected to a temperature of 269 degrees Celsius, which is the precise temperature at which the pure L-threonine crystal starts the process of melting.

The connection between hardness number and load for pure and 0.5 mol% dye doped L-threonine materials, which have been cooled slowly at different temperatures, is shown in figures 17 and 18. Based on the graph, the H_v value exhibits a positive correlation with the load and a negative correlation with the temperature up to 303 K.

Temperature intensification induces matrix (lattice) vibration, leading to an increase in the constant interatomic spacing compared to that seen at normal temperature. The interaction with an additional lattice phonon leads to a breach of the bond and a decrease in the hardness rate. Currently, it is possible to manufacture NLO-ground devices with temperatures reaching 269 °C for pure L-threonine and 270 °C for L-threonine doped with 0.5 mol% of dye, by using crystals of dye-doped L-threonine. Furthermore, the data presented indicates the Vickers microhardness values for completely doped crystals at room temperature.



3.13. Half voltage

The voltage half-wave V_{π} should be minimised and is necessary for a mandatory level of suppression, which is a crucial factor to consider when choosing a cryogenic sample. The mechanical contact demonstrates excellent dielectric breakdown, enhanced optical resolution in the surrounding region, and great ohmic contact. In a transverse design, the half-wave voltage of a motor varies inversely with the aspect ratio L . Half-wave signals exhibit voltage values that span from tens of volts to 10,000 volts when used with long-term devices. Nevertheless, the transverse dimension 'd' is limited by both the diffraction of the small cross-sectional light beam and the capacitance of the semiconductor. The electro-optic molecular sampler setup consists of a laser, a high-quality crystal sample, an analyser, a photo detector, a polarizer, and other necessary components, in

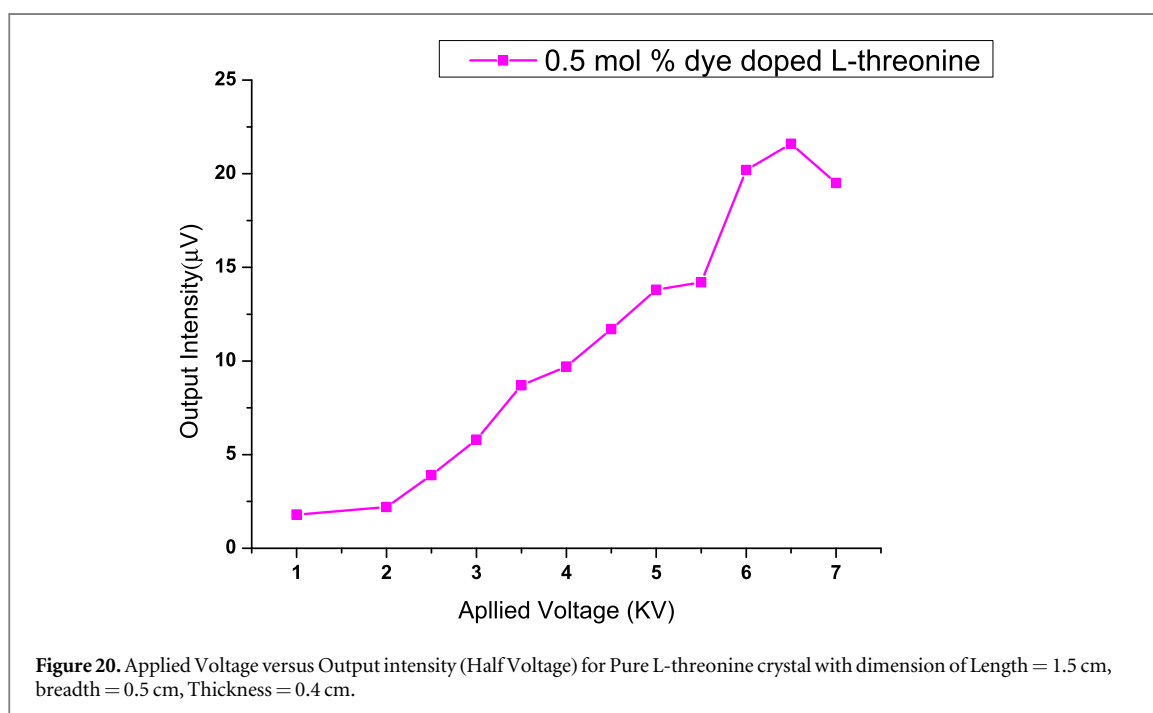
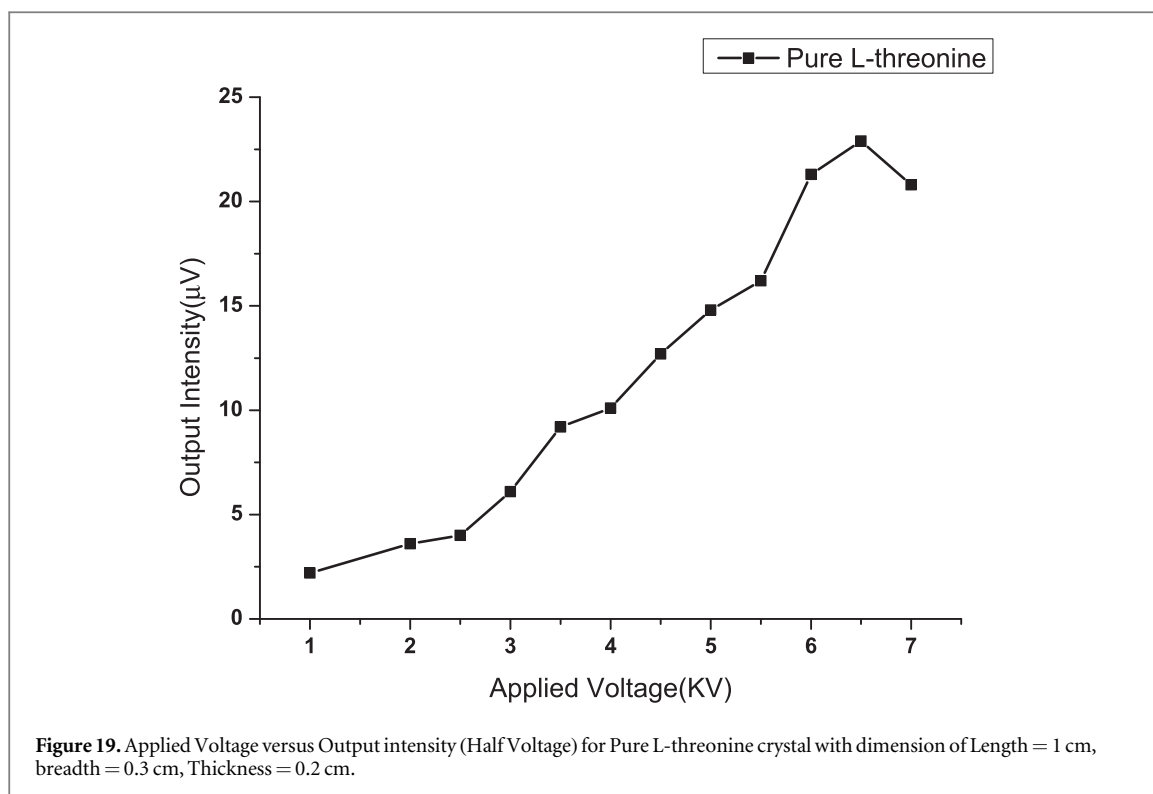
Table 7. The statistics of functional voltage versus output strength for Pure L-threonine crystal specimen with dimension of Length = 1 cm, breadth = 0.3 cm, Thickness = 0.2 cm.

Input (kV)	Output (μ V)
1	2.2
2	3.6
2.5	4
3	6.1
3.5	9.2
4	10.1
4.5	12.7
5	14.8
5.5	16.2
6	21.3
6.5	22.9
7	20.8

Table 8. The statistics of functional voltage versus output strength for 0.5 mol % dye doped L-threonine crystal specimen with dimension of Length = 1.4 cm, breadth = 0.5 cm, Thickness = 0.4 cm.

Input (kV)	Output (μ V)
1	1.8
2	2.2
2.5	3.9
3	5.8
3.5	8.7
4	9.7
4.5	11.7
5	13.8
5.5	14.2
6	20.2
6.5	21.6
7	19.5

addition to the voltage originator. Furthermore, the significance of the lower half wave voltage of Pure L-threonine and dye doped L-threonine coefficients is evident in their role throughout the design and production phases of the electronic modulator. In the laboratory, the fusion of the electro-optic modulator-driver result in a drop in the half wave voltage by using the ratio between the reciprocal and geometric aspect ratios. Tables 7 and 8 show the relationship between the provided volatility and the output strength for different sizes of crystals measured with a detector. A correlation was established between the applied voltage and the output power by analysing the voltage spikes detected. Figures 19 and 20 displayed the measurement of the half wave voltage, which was obtained by comparing the applied voltage with the peak strength of the output. The highest value received from the group was interpreted as the voltage at which the half wave occurs, known as the V_{π} voltage. The half-wave voltage be determined using the formulae for transverse phase modulators as follows: The equation $V_{\pi} = \frac{\lambda d}{nx^3rL}$ gives the value of the half-wave voltage. The value of V_{π} for the longitudinal phase modulator is given by $V_{\pi} = \frac{\lambda}{nx^3r}$. The analysis revealed that the highest point on the graph matched to the voltage known as V_{π} , which is the half-wave voltage. The measurement was conducted outdoors at a temperature of 33 degrees Celsius. The half-wave voltage is the inverse of the electro-optic coefficient [27]. Pure L-threonine crystals have a lower half-wave voltage compared to L-threonine crystals doped with dye. The half-wave voltage values show a drop when dye is doped into the pristine material. In the laboratory, the half-wave voltage for the hypothetical electro-optic modulation be reduced by increasing the quality factor and achieving a desirable element ratio.



3.14. Contact angle

Contact angles were measured to study the wettability of dye-doped and pure L-threonine on the (2 1 0) plane. Honey served as the experimental liquid medium for this investigation using a digital camera with a 30x optical zoom. The picture was obtained via magnification, showcasing the serpent's construction. The contact angle data was obtained from the picture using the Java programme ImageJ. Honey droplets were positioned on the (2 1 0) plane, and mean contact angle measurements were conducted to get a more precise result. Table 9 clearly demonstrates that the contact angle rises for dye-doped L-threonine samples in comparison to pure L-threonine samples. The enhanced thermal stability of dye-doped L-threonine crystals is also verified by the introduction of methylene blue molecules into L-threonine. The thermal analysis of both doped and undoped L-threonine crystals confirmed the occurrence of a structural phase change at elevated temperatures. This transition was seen

Table 9. Mean contact angle honey droplets on pure and dye doped L-threonine.

Crystal	Contact angle in degree
Pure L-threonine	58.5 ± 0.5
0.5 mol % dye doped L-threonine	62.4 ± 0.2

by measuring the lowered surface tension and enhanced hydrophilic properties of L-threonine specimens in the presence of honey. Moreover, the observed average contact angle in dye-doped L-threonine rapidly rose due to decreased tension in the (2 1 0) growth sectors and the connection of dye-methylene blue molecules to the ionic assembly of the L-threonine crystal. Furthermore, it demonstrates the ability of the amalgam to attach to the lipid membrane and narrow down the metal-binding domain of the bacterial enzymes.

The surface energy, surface roughness, and surface chemistry of the solid material are some of the variables that could impact the contact angle of a liquid on a solid surface. A higher contact angle between a pure L-threonine crystal and one doped with methylene blue dye could arise from a number of factors: incorporating methylene blue dye molecules into the crystal structure modifies the surface chemistry of the crystal. This modification has the potential to cause alterations in the interaction between the surface of the crystal and the liquid, which lead to an elevation in the contact angle. The introduction of dye molecules into a crystal by doping has the potential to cause abnormalities or alterations in the surface roughness. Heightened surface roughness influences the wetting characteristics of the crystal, resulting in an elevation of the contact angle. Dye molecules can alter the surface energy of the crystal. The contact angle be influenced by the change in surface energy, which is determined by the characteristics of the dye and its interaction with the crystal surface. Methylene blue and other dye molecules have the ability to establish precise contacts with the crystal surface, including hydrogen bonding and π - π interactions. The interactions between molecules affect the ability of the crystal to be wetted and can lead to an increase in the contact angle.

3.15. Piezoelectric study

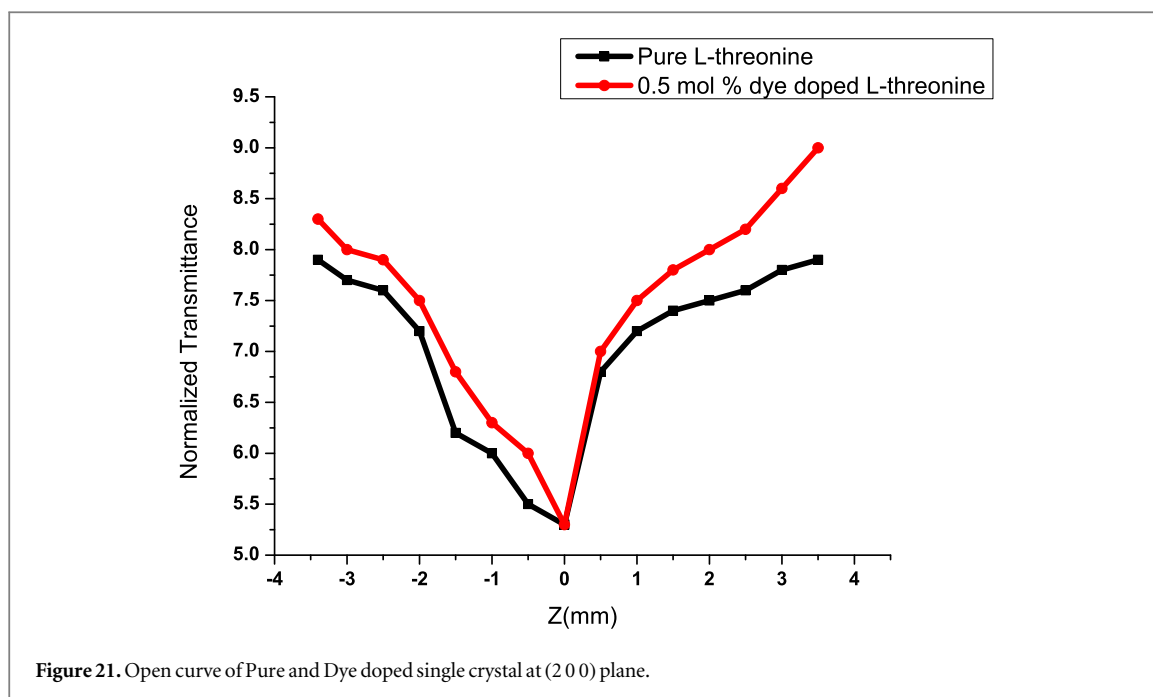
Non-centrosymmetric single crystals often exhibit a range of advantageous features, such as piezoelectricity. Piezoelectricity refers to the phenomenon of electric charges accumulating in crystalline materials as a result of mechanical strain. Crystals without a core of symmetry often exhibit piezoelectric behaviour. Centrosymmetric crystals prevent the existence of a dipole moment due to the cancellation of ion motion by the material's symmetry. Typically, materials that lack a centre of symmetry and have a high capacity to be polarised tend to have a higher piezoelectric coefficient. The crystals underwent meticulous polishing and were subjected to a voltage of 1 kilovolt per millimetre for a duration of 3 h. The piezoelectric coefficient d_{33} was determined to be 1.92 pc/N for pure L-threonine and 2.76 pc/N for dye-doped L-threonine. The increase in coefficient is caused by the creation of an electric dipole resulting from an imbalance of charges when stress is exerted.

The addition of dye to L-threonine crystals has been shown to be a very effective dopant, making it suitable for a wide range of applications such as transducers, sensors, and actuators.

3.16. LTD

In order to be suitable for non-linear optical applications, a material must possess a high threshold for laser damage. This is due to the fact that the effectiveness of harmonic conversion is directly correlated with the strength of the incoming beam. The maximum allowable power of a high-intensity laser potentially causes substantial damage to the sample. The laser damage threshold is a crucial parameter for nonlinear optical applications.

Laser-induced damage occurs on the surface of the sample when it is subjected to a 532 nm Nd:YAG laser with a pulse width of 7 ns and a repetition rate of 10 Hz. A convex lens was placed precisely at the focus point of the laser beam in order to direct it towards the crystal. The laser pulse's energy was measured using a phototube and an oscilloscope, while an attenuator was used to modify the laser's energy. The formula for power density (P_d) = $\frac{E}{\tau \pi r^2}$, where E is the input energy (mJ), τ is the pulse width, and r is the beam radius, was used to calculate the power density. The laser beams speck diameter is 1 mm, and its energy was increased from 5 mJ. The energy threshold at which the laser becomes damaged was determined by measuring the energy of the laser beam using an energy metre. Crystals of Pristine and dye-infused L-threonine were chosen for evaluation of their laser damage threshold, with a consistent thickness. The beam was transmitted via the (2 0 0) crystal faces. The laser damage threshold of Pristine L-threonine was determined to be 25.6 mJ cm⁻², whereas for Dye integrated L-threonine it was measured to be 33.4 mJ cm⁻². The micrographs of LDT pattern are shown in figure 23. The presence of a greater number of dislocations in Dye integrated L-threonine crystal has a significant impact on the damage threshold value, which is higher compared to Pure L-threonine.



Methylene blue dye molecules are more effective in absorbing and dissipating laser energy than pure L-threonine. The dye molecules absorb some of the laser energy, which slows the L-threonine crystal's advancement towards the damage threshold. Doping L-threonine with methylene blue may enhance its nonlinear optical properties. The dye molecules might promote more efficient laser energy dissipation and help nonlinear optical processes by creating new electronic states and transitions. Methylene blue molecules may improve the quality of the crystal by reducing defects or increasing the regularity of the crystal lattice. When there are fewer defects, there are fewer weak points where laser damage might occur. The dye molecules may aid in energy transfer mechanisms that distribute the acquired energy more evenly throughout the crystal. By avoiding localised heating and damage, this might increase the damage threshold overall.

Methylene blue may photochemically stabilise the L-threonine crystals when exposed to laser light, reducing the likelihood of bond breaking or other forms of degradation. In conclusion, by increasing the crystal's ability to absorb, distribute, and transmit laser energy, doping L-threonine with methylene blue dye elevates the laser damage threshold over that of pure L-threonine.

3.17. Etching

The optical quality of harvested crystals be assessed by examining the defects using chemical etching procedures. The primary criterion for the etching inspection is that the crystal must possess optical clarity and must be free from any microcracks or surface contaminants. The crystals should be only examined on their exposed surfaces. To prepare the surfaces for etching, immerse the plates in etchant (water) for 20 s and then wipe them off using suitable paper. The experiments were carried out on L-threonine single crystals that included both Pure and dye doped. The etch pit density of pure L-threonine was measured to be $2.8 \times 10^2 \text{ cm}^{-2}$, whereas the etch pit density of L-threonine with dye incorporation was measured to be $1.6 \times 10^2 \text{ cm}^{-2}$. The results indicate that L-threonine crystals containing dye have a lower number of dislocations compared to pure L-threonine crystals. This is also consistent with contact angle analysis. Lower surface energy leads to reduced dislocation density, resulting in fewer step generations compared to higher energies. Therefore, the research shows a significant increase in the crystalline structure of L-threonine when dye is introduced, compared to the original L-threonine.

3.18. Z-Scan

The third order Z-Scan measurements were used to examine the nonlinear optical characteristics of Pure and dye doped L-threonine. This method, which is one of the most popular ones, makes it possible to measure a non-linear absorption coefficient and a non-linear refractive index simultaneously in terms of both magnitude and sign (n_2). Mathematical equations are used to determine the susceptibility of the Z-Scan NLO for the imaginary and real parts of values [28]. He-Ne laser with an operating wavelength of 632.8 nm be used to perform open and closed aperture Z scan experiments.

From the collected data of the Z-Scan study disclosure the linearity in prominent plane (2 0 0) of dye doped L-threonine is higher than pure L-threonine plane (2 0 0) depicted in figures 21 and 22. The crystal specimen's

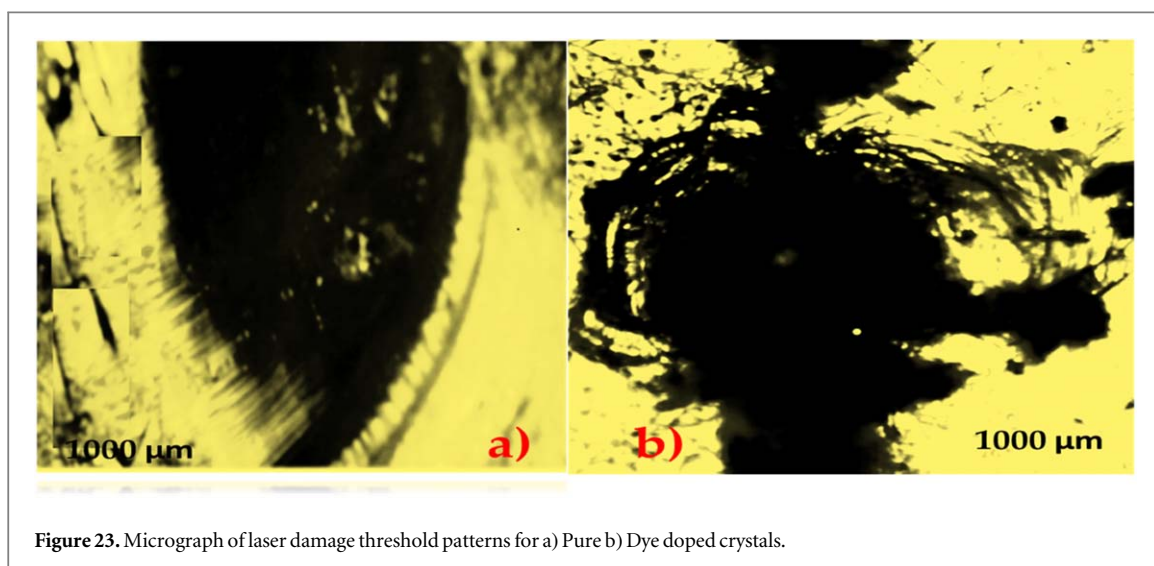
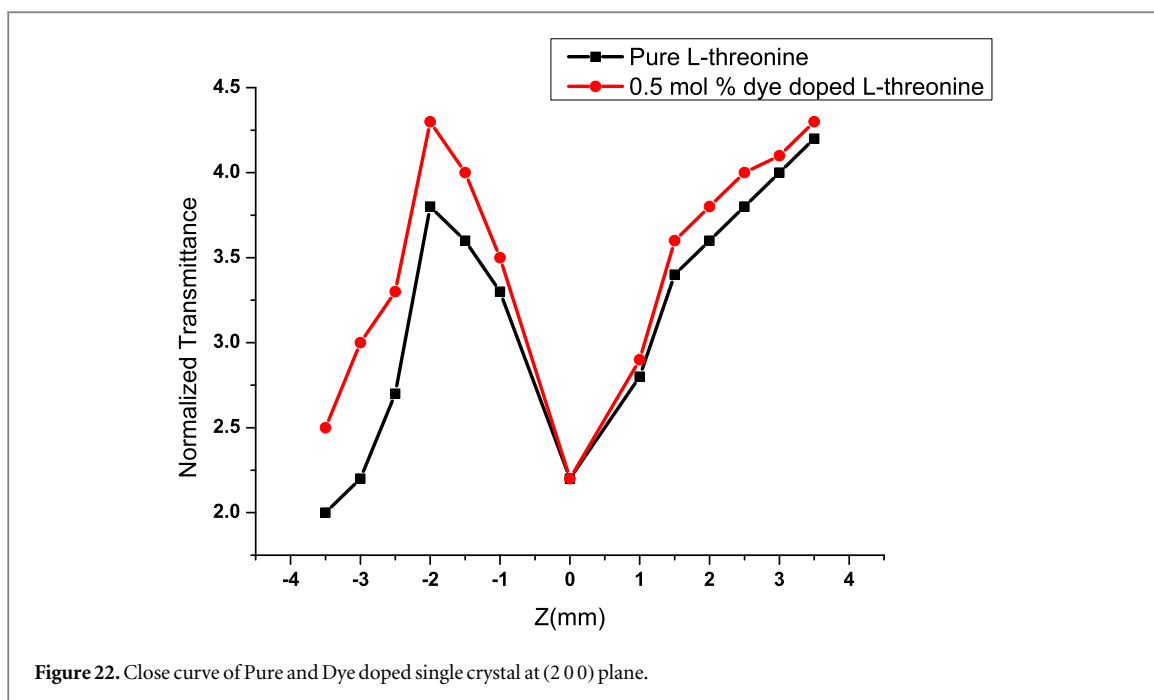


Table 10. Comparison of Z-Scan values of Pure and dye doped L-threonine.

Crystal	Third order optical nonlinear susceptibility $\chi^{(3)}$ (esu)
Dye doped L-threonine (present work)	8.3452×10^{-5}
Pure L-threonine	5.90×10^{-6}

self-defocusing characteristic is a consequence of a multi-photon absorption approach, leading to nonlinear refraction and absorption. Moreover, values are shown in table 10. The dimethyl amino group of methylene blue molecules admixture in dye doped L-threonine, leading to a third order optical nonlinear susceptibility that is 8.3452×10^{-5} (esu) higher than that of pure Crystal.

4. Conclusion

The Standard slow evaporation process was utilised to harness the isolated crystals of Pure and Dye doped L-threonine. X-ray diffraction analysis provides insights into the anisotropy and phase purity of the material.

The FTIR measurement confirms the presence of functional groups in the substance. The UV–vis investigation showcases the material's capabilities to be used in photonic applications that selectively manipulate wavelengths. The Powder XRD was used to identify the major peaks, and the presence of additives in the parent matrices was confirmed using SEM-EDX analysis, which also allowed for the quantification of components. The use of dopants in the mechanical hardness test results in an enhanced mechanical stability. The laser damage threshold exhibited a higher value in comparison to Pure L-threonine. The presence of dye enhances the piezoelectric coefficient. The contact angle exhibited a linear rise in L-threonine crystals that had been infused with dye. In present work, we are focussed on methylene blue, an organic dye as a dopant for increasing high efficiency third order non-linearity value in crystals. A SHG study depicts that additives have enhanced the molecular alignment of the L-threonine, raising the NLO and enhancing the production of opto-electric devices. The effect of dye reduces bandgap leads to predominant over wavelength -selective photonic application. Temperature dependence hardness study of pure and dye doped L-threonine claims its resistance over plastic deformation. SHG study over different particle size focusses on microscopic enhancement of Second harmonic generation.

The nonlinear characteristics of the material were examined by Kurtz Perry and Z-Scan analysis. Particularly concentrating on dye-incorporated crystals, the third order nonlinearity for both pure and dye-doped L-threonine was observed.

Data availability statement

The data cannot be made publicly available upon publication because they contain commercially sensitive information. The data that support the findings of this study are available upon reasonable request from the authors.

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